
An Aggregate B-to-C Lower Bound from Revealed-Sacrifice Observation

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Abstract

The Goal-Frontier Maximization framework’s central measure vol_P quantifies capability *possession*, but the experiential value it protects requires capability *exercise* (vol_R). Global direct measurement of vol_R is privacy-incompatible at scale. We introduce *revealed-sacrifice observation*: under assumptions S0–S2, S4(a) (S1 = free choice with non-degrading alternatives), and the genuine-exchange conditions of Definition 2, each voluntary trade event—an agent surrendering a benchmarked X for an unbenchmarked bundle Y —yields a lower bound $\text{vol}_R^{[W]}(Y) \geq \Delta \text{vol}_P(X)$ on Y ’s exercise contribution within window W . Aggregating across events whose bundles are individually cooperative-closed and pairwise poset-independent yields the *Aggregate B-to-C Lower Bound theorem*: on the observed subset U , $\text{vol}_R^{U,[W]} \geq \sum_n \Delta \text{vol}_P(X_n)$. The bound is non-decreasing in observed events and composes with the commitment protocol of [Lasser, 2026a, Exogenous Verification] via two interfaces (per-event and hiding-variant rollout) both with three formal privacy properties plus two named institutional preconditions (Proposition 1). Money and time sacrifice channels triangulate the same underlying substrate. We also show that vol_R ’s mapping from capability systems is not strictly monotone under removal of unexercised capabilities, so vol_P ’s self-balancing property does not transfer to vol_R ; vol_R is a diagnostic, not an optimization target. The need-sufficiency S1 filter, gap decomposition, wireheading-consistent concentration signal, and residual class are developed in [Lasser, 2026d, Need Sufficiency].

1 Introduction

The GFM framework defines its actor as a system that maximizes vol_P , the poset measure over the capability space [Lasser, 2026b, Definition 6]. The axiomatization of [Lasser, 2026f, Proposition 1] establishes vol_P as a poset measure satisfying axioms M1–M6 and proves it self-balancing [Lasser, 2026f, Theorem 1]. The analysis of [Lasser, 2026c, Horizon Aware] extends to multi-substrate collectives, proving that full domination is anti-maximizing under discounting [Lasser, 2026c, Proposition 6]. Throughout, vol_P is treated as a faithful proxy for the experiential optionality the framework is meant to protect.

But vol_P measures what the collective *can* do, not what it *does*. The B-to-C gap, named in [Lasser, 2026b, Section 7] and flagged as a primary open problem in the sequence, is the possibility that these diverge. Two failure modes illustrate:

1. **The Doll Problem** [Lasser, 2026b, Section 7.1]: a collective possesses rich capabilities but exercises few. vol_P is high but experiential value is zero.
2. **Wireheading** [Lasser, 2026a, Discussion]: a collective exercises capabilities in narrow self-reinforcing loops that serve the measurement system rather than experiential value. High vol_P and apparent exercise, but the exercise is degenerate.

Both are instances of Goodhart’s Law [Strathern, 1997, Manheim and Garrabrant, 2019]: when the measure (vol_P) diverges from the target (experiential optionality), optimizing the measure no longer serves the target.

1.1 Why direct measurement fails

A natural first approach is to define a *realized capability volume* vol_R by restricting vol_P to an exercised sub-poset—capabilities counterfactually load-bearing for realized outputs—and monitor the ratio $\beta = \text{vol}_R / \text{vol}_P$. This fails on two grounds:

Tractability concern. The counterfactual query (“which capabilities are load-bearing for which realized outputs?”) requires global knowledge of every realized output the collective produces and of every causal pathway from each capability to those outputs. Establishing a complexity lower bound is outside this paper’s scope; the qualitative observation that direct computation faces high practical cost at collective scale, alongside the privacy concern below, motivates the indirect channel.

Privacy failure. Computing realized-output sets requires observational access to what the collective actually does. This is the total-surveillance scenario. The framework’s operational commitments [Lasser, 2026a,e,h] rule out panopticon-style observation.

1.2 What this paper does

We observe that agents *voluntarily disclose* vol_R -content through an existing, privacy-respecting channel: **sacrifice**. When an agent surrenders a benchmarkable capability (whose vol_P contribution is known) in exchange for an unbenchmarked bundle (whose vol_R contribution is unknown), S1 (free choice with non-degrading alternatives) makes the trade voluntary, and revealed preference establishes a valuation comparison ($U_i(Y) \geq U_i(X)$); composing this with S0 (valuation-bounded-by-exercise), S2 (benchmark grounding), and S4(a) (valuation additivity) under the genuine-exchange conditions of Definition 2 yields a lower bound $\text{vol}_R^{[W]}(Y) \geq \Delta \text{vol}_P(X)$ on the unbenchmarked bundle’s exercise contribution. Aggregating across events with pairwise poset-independent bundles produces a constructive lower bound on vol_R on the observed subset U , equivalently an upper bound on the subset-local B-to-C gap $1 - \beta^{\text{lower}}(U, T)$ (with $\text{vol}_P(U)$ in the denominator), or equivalently on the global conservative gap $1 - \beta^{\text{lower}}(P, T)$ when the denominator is $\text{vol}_P(P)$ and $\text{vol}_R^{\text{lower}}$ is zero-extended outside U (Definition 6).

This paper formalizes the sacrifice event (Section 2), states the six assumptions (S0–S5) under which the aggregate bound holds, proves the central theorem (Section 3), analyzes the two observation modalities (money and time, Section 4), establishes the structural non-self-balancing finding for vol_R (Section 5), and composes the channel with the commitment protocol of [Lasser, 2026a, Exogenous Verification] (Section 6).

1.3 Scope: what this paper does not do

- **It does not filter S1-failing below-sufficiency sacrifices.** Not every observed sacrifice event is voluntary—agents pay below-sufficiency costs (dehydration, debt-servicing) when no non-degrading alternative is available, in which case S1 fails (retention is degrading and no third non-degrading option exists). Such events have reversed polarity relative to the theorem’s assumptions. The companion paper [Lasser, 2026d, Need Sufficiency] introduces the *need-sufficiency architecture* that supplies the structural condition for diagnosing this S1 failure mode; the filtering itself remains an institutional-attestation step. Below-sufficiency targeting where a third non-degrading alternative is available may still satisfy S1; classification of such events likewise routes through institutional attestation. The present paper takes the filter as an external precondition: events reaching the aggregate are assumed S1-admissible.
- **It does not decompose the B-to-C gap into diagnostic categories.** Interpreting why $\beta^{\text{lower}} = \text{vol}_R^{\text{lower}} / \text{vol}_P$ is low requires partitioning capabilities into restricted, covered, dormant, and residual cells and tracking which cell dominates the gap. The gap decomposition machinery and its wireheading-consistent concentration signal are developed in [Lasser, 2026d, Need Sufficiency].

- **It does not provide a complete theory of experiential value.** The residual class (capabilities structurally outside the sacrifice channel’s reach) is characterized in [Lasser, 2026d, Need Sufficiency].
- **It does not replace vol_P with vol_R .** The framework continues to optimize vol_P ; the revealed-sacrifice lower bound is a diagnostic that detects when vol_P optimization has gone wrong. vol_R is the audit, not the objective.
- **It does not measure vol_R directly.** The paper produces a *lower bound* on vol_R from trade events, not a precise measurement. An *exact* per-capability exercise witness $\text{vol}_R^{\text{exact}}$ within consented perimeters, together with the boundary-closure conditions and OI-exercise bridge that license combining $\text{vol}_R^{\text{exact}}$ with the bound’s $\text{vol}_R^{\text{lower}}$ outside the perimeter into a unified diagnostic $\text{vol}_R^{\text{combined}}$, is developed in Appendix A. The combined diagnostic is the natural deployment object for governance frameworks that operate both inside and outside consented perimeters; main-text sections 3–5 state results for the privacy-respecting lower-bound channel alone, and forward-point to the appendix where the combined-diagnostic infrastructure is needed.

1.4 Summary of results

1. **Privacy-respecting observation channel with formal-plus-institutional privacy split** (Section 2). Revealed-sacrifice observation defines the sacrifice event and states six assumptions (S0–S5). Three formal privacy properties are established (Theorem 1: no interior access, emission-rate discretization—bounded by trade frequency rather than observation rate, not a per-event information-leakage bound, commitment-layer composability); two further properties hold operationally only under named institutional preconditions (Proposition 1: consent via participation, third-party observability). The “privacy-minimal” interpretation is conditional on the institutional preconditions; the formal layer alone establishes P1, P3, P4.
2. **Aggregate B-to-C Lower Bound** (Section 3). Theorem 2 lower-bounds $\text{vol}_R^{U,[W]}$ —the window-active realized capability volume on the subset U covered by S1-admissible trade events (free-choice voluntary trades with at least one non-degrading alternative in the choice set) with *pairwise poset-independent bundles*—by aggregated sacrifice magnitudes. Via Definition 6, the bound induces two associated ratios: the subset-local exercised fraction $\beta^{\text{lower}}(U, T) = \text{vol}_R^{\text{lower}}(U, T)/\text{vol}_P(U)$ on the observed U , and the global conservative fraction $\beta^{\text{lower}}(P, T) = \text{vol}_R^{\text{lower}}(U, T)/\text{vol}_P(P)$ on the full capability-space denominator. The subset-local ratio is the lower bound restricted to U ; the global ratio is a lower bound on the true whole-system exercised fraction. Both are lower bounds, not tight estimates: no sharpness/optimality theorem is claimed. We write simply β^{lower} when the choice of denominator is fixed by context. Part A (bundle-level, requires S0–S2 and S4(a)) is the main result; Part B (per-capability, requires S0–S5 plus within-bundle poset-independence) is a conditional refinement. The constructed lower bound is non-decreasing in observed admissible events: Part A via sub-batch supremum (Proposition 3) and Part B via per-capability max-attribution (Proposition 2).
3. **Two sacrifice channels, substitutable with residuals** (Section 4). Money-sacrifice and time-sacrifice are substitutable via cooperative markets (employment converts time to money; hiring converts money to time). The channels triangulate the same underlying substrate rather than covering disjoint capability spaces. Non-fungibility residuals (agent-specific time, capital-concentrated access) are existential witnesses enabling strict improvement of the joint aggregate conditional on observed-coverage content (Corollary 1).
4. **Non-self-balancing finding** (Section 5). vol_R is non-invariant under capability removal: unexercised capabilities can be eliminated without reducing vol_R , while vol_P contracts by their weight. The self-balancing property therefore does not transfer from vol_P to vol_R , so vol_R is a diagnostic, not an objective. This structural finding is independent of the observation mechanism.
5. **Commitment-layer composition** (Section 6). The sacrifice channel composes with the commitment protocol of [Lasser, 2026a, Exogenous Verification] to produce a privacy-minimal aggregate on distinct-category batches (Propositions 4 and 5). The rollout ZK-

certifies the Part A formula inputs; the theorem’s non-ZK hypotheses require external attestation (Remark 18).

1.5 Design principle

This paper shares a structural discipline with [Lasser, 2026e, Phase Redundancy] and [Lasser, 2026h, Controlled Relaxation]: **resolution through frequency, not through depth**. Controlled relaxation [Lasser, 2026h, Controlled Relaxation] gains precision by running more tests, not by surveilling each test more intrusively. Revealed-sacrifice observation gains precision by observing more trades, not by observing each trader more intrusively.

1.6 Dependencies on prior papers

A table summarising the prior-paper results this paper relies on is collected in Appendix C. The companion paper [Lasser, 2026d, Need Sufficiency] and the earlier sequence papers [Lasser, 2026b,f,c,g,a,e,h] are presently distributed as preprints. The propositions and definitions imported here are pinned by their current formulations; if the upstream preprints revise their numbering or content, downstream references will need refreshing correspondingly. Stable archival identifiers will be added as the sequence reaches archival status.

2 Revealed-Sacrifice Observation Model

The privacy-compatible observation channel for vol_R is *revealed sacrifice*: agents disclose the value they place on unbenchmarked capabilities through the benchmarked capabilities they voluntarily surrender to obtain them. The framework does not observe *what* the agent values privately, only *what they commit to publicly through sacrifice*.

2.1 Assumptions

Notation used throughout this section. A sacrifice event is denoted as a tuple (i_n, X_n, Y_n, t_n) with i_n the agent, X_n the surrendered benchmarked capability (or bundle), Y_n the acquired unbenchmarked bundle, and t_n the event timestamp; the formal definition with its genuine-exchange conditions appears as Definition 2 in Section 2.2. We use the tuple notation in the assumption statements below.

The revealed-sacrifice channel and its aggregate bound (Section 3) operate under six named assumptions. We collect them here for single-point reference.

S0 (Valuation-bounded-by-Exercise). For each agent i_n with event at $t_n \in W$, the agent’s valuation of the unbenchmarked bundle Y_n does not exceed Y_n ’s *window-active* realized vol_R contribution: $U_{i_n}(Y_n) \leq \text{vol}_R^{[W]}(Y_n)$. This is the *substantive bridge* between valuation (what the agent pays for at t_n) and realized exercise (what the framework cares about, aggregated over the observation window W). S0 is a per-event one-sided conservatism claim: for each individual event n , the agent’s valuation does not exceed the bundle’s W -window exercise contribution. The empirical rationale is that typical agents in typical markets do not pay more than what the capability delivers; this rationale supports adopting S0 but is not itself part of the per-event statement. The window-active form is essential: the sacrifice channel observes acquisition at t_n , not exercise at any particular time, so S0’s consequent is about the cumulative exercise witness across the window rather than the instantaneous value at t_n . Agents who buy capabilities they never use within W violate S0 for those events; the lower bound is about *realized* rather than *stored* capability.

S1 (Free choice). Each sacrifice event is a voluntary trade: agent i_n chose Y_n over X_n when both were available, the trade was free of *material confounds* (no transaction costs that would distort the comparison beyond $\Delta \text{vol}_P(X_n)$, no hidden side payments altering the effective X or Y , no risk/expectation mismatch between $U_{i_n}(Y_n)$ at trade time and the realized $\text{vol}_R^{[W]}(Y_n)$, and no unmodeled bundle attributes outside the declared Y_n that contribute to valuation), and the agent had at least one *non-degrading* alternative in their choice set: an option (retaining X_n , or some third option not equal to the trade itself) that would

not have left the agent *below-sufficiency* on any need dimension. For this paper we treat “below-sufficiency” as a primitive: an event is below-sufficiency iff retaining X_n would leave the agent unable to function on at least one need dimension without corrective intervention (food, water, shelter, medical care, etc.). S1 fails for events in which *no* non-degrading alternative exists in the agent’s choice set (in particular: retention is degrading *and* no third non-degrading option is available) or in which the material-confound conditions are violated; the institutional attestation layer (Remark 18) is responsible for filtering such events from the aggregate. S1 holds (and the event is admissible) when retention is non-degrading, or when retention is degrading but the agent has a non-trade non-degrading alternative; the latter case requires attestation that the third alternative was actually available and non-degrading. The full poset-construction machinery that identifies below-sufficiency structurally (multi-dimensional need bundles, downstream-safe sufficiency, polarity-correct benchmarks, and the downstream-gated weighting that makes need-satisfaction a precondition for vol_P -registration of downstream capability weight) is developed in the companion paper [Lasser, 2026d, Need Sufficiency]. This paper does not depend on that machinery for any proof; the companion paper operationalizes what this paper axiomatizes as a primitive.

S2 (Benchmark grounding). The benchmarked sacrifice X_n has a well-defined vol_P contribution $\Delta \text{vol}_P(X_n)$ computed via [Lasser, 2026f, Proposition 1], and the agent’s valuation of X_n is at least as large: $U_{i_n}(X_n) \geq \Delta \text{vol}_P(X_n)$.

S3 (Bundle coherence). The hedonic decomposition of each bundle Y_n into capabilities in U is well-defined (no multicollinearity degeneracy).

S4 (Additive separability).

- (a) *Valuation additivity (global):* U_i is additive over any finite capability set: $U_i(S) = \sum_{c \in S} u_i(c)$ for any finite $S \subseteq P$, with $u_i(c) \geq 0$. In particular this applies to sacrifice bundles ($U_i(X) = \sum_{c \in X} u_i(c)$), acquired bundles ($U_i(Y)$), and agent holdings ($U_i(H)$). Complementarities are handled at the poset level by treating cooperative capabilities as their own nodes (M6), which appear as additional capabilities in S whose u_i accounts for the cooperative term.
- (b) *vol_R additivity on poset-independent capabilities:* The vol_R contribution of a set of capabilities that are pairwise poset-independent (no subsumption relations *and* no cooperative composition links) is additive.

S5 (Decomposition validity). For events with $\text{vol}_R(Y_n) > 0$, the event-local decomposition coefficients are *lower bounds* on the true vol_R share: $\alpha_{n,c} \leq \text{vol}_R(c)/\text{vol}_R(Y_n)$ for all such n and $c \in Y_n$. Events with $\text{vol}_R(Y_n) = 0$ carry no share information and are excluded from Part B by the positivity precondition of the proof. Under the Part B hypothesis of within-bundle poset-independence (Definition 4), $\sum_{c \in Y_n} \text{vol}_R(c)/\text{vol}_R(Y_n) = 1$, so the share lower bounds and the partition of unity $\sum_c \alpha_{n,c} = 1$ jointly force $\alpha_{n,c} = \text{vol}_R(c)/\text{vol}_R(Y_n)$ exactly: under Part B’s hypotheses, S5 collapses to a share-equality assumption rather than a strict inequality. We retain the inequality form because it is the operationally natural default—practitioners typically have lower-bound share estimates rather than exact share knowledge—but acknowledge that under within-bundle independence the inequality is binding throughout, so Part B is effectively conditional on the analyst supplying *exact* share estimates compatible with the true (unknown) shares.

Remark 1 (S5 is a conditional-refinement assumption). *S5 is the strongest of the six assumptions: it asserts a direct share inequality on the quantity Part B of Theorem 2 bounds. Part A makes no per-capability attribution claim and does not require S5. Part B’s per-capability refinement should be understood as a conditional result: under the additional structural assumption that decomposition coefficients are conservative share lower bounds, and the hypothesis of within-bundle poset-independence (Definition 4, Remark 9), the aggregate can be refined to per-capability bounds. Within-bundle poset-independence is what makes S5’s share inequality compatible with the partition-of-unity constraint on the α coefficients: bundles with internal cooperative activation produce $\text{vol}_R(Y_n) > \sum_{c \in Y_n} \text{vol}_R(c)$, forcing $\sum_{c \in Y_n} \alpha_{n,c} < 1$ under S5 and violating Definition 4. Combined with the share-collapse note above, this means Part B’s strict admissibility requires the analyst to supply $\alpha_{n,c}$ values equal to the true (unknown) vol_R shares, not merely lower bounds. In practice the true shares are not known precisely; Part B is therefore operationally available only when the analyst can establish either roughly-homogeneous bundle structure (where defaults are*

close to true shares) or an externally-attested share estimate. S5 is a substantive empirical assumption, not automatically satisfied by any single default. The equal-weight default $\alpha_{n,c} = 1/|Y_n|$ is not a generic conservative choice: it satisfies S5 only when the true vol_R shares are each at least $1/|Y_n|$, which fails whenever any capability in the bundle dominates the others. Operational settings where S5 plausibly holds include: (a) bundles known to be roughly homogeneous in vol_R contribution (the equal-weight default is then close to the true shares), (b) explicitly conservative attributions where $\alpha_{n,c}$ is set strictly below any plausible share estimate from prior hedonic regressions (Rosen, 1974) or domain analysis, and (c) bundle configurations where one capability is known a priori to dominate and is given an attribution at most that capability's true share. Each of these is a context-specific assumption, not a universal property of the decomposition.

Part A of Theorem 2 requires S0–S2 and S4(a): S4(a) is the valuation-additivity assumption used in the exchange-to-bundle reduction of the revealed-preference inequality (Step 2 of the justification in Section 2.2). Part B additionally requires S3 (bundle coherence), S4(b) (vol_R additivity on poset-independent capabilities), S5 (decomposition validity), and within-bundle poset-independence of each admissible Y_n (Definition 4). The need-sufficiency architecture of the companion paper [Lasser, 2026d, Need Sufficiency] characterizes when S1 fails structurally and provides the filter that keeps the sacrifice channel clean; this paper treats the filter as an external precondition supplied by Remark 18.

Remark 2 (S0 is the substantive bridge, not a calibration detail). *S0 does the conceptual work of the theorem. Revealed-preference reasoning on its own establishes only that traded bundles are valued at least as highly as surrendered bundles; it does not establish that what the agent values corresponds to what the capability delivers in exercise. S0 asserts precisely that correspondence (in the one-sided direction needed for a lower bound): subjective valuation is upper-bounded by realized exercise contribution. This is a non-trivial behavioural claim. Agents who systematically overpay for status, novelty, or anticipated-but-unrealized utility violate S0 in realistic markets; the theorem's empirical validity depends on the prevalence of such violations.*

S0 does not require that agent valuations perfectly track vol_R , only that they do not overstate it. Agents may undervalue unbenchmarked bundles (sacrifice little for something very valuable), in which case the lower bound is conservative but still valid. S0 fails only in the overpayment direction. Weakening S0 to expected/distributional conservatism (Open Question 7) is a priority direction for follow-up work.

Systematic overpayment patterns. In markets with status goods, speculative assets, or anticipation-driven purchases, agents routinely sacrifice substantially more than the exercise value they will realize within any reasonable window. Ritualized payment (sacrifices made for participation rather than benefit), gift aversion (sacrifices that signal commitment beyond exercise contribution), and time-for-money inefficiency (sacrifice-rate calibration that diverges from exercise contribution because of preference-misalignment with the wage channel) are all empirically prevalent failure modes of S0. The theorem's empirical validity is bounded by the prevalence of such violations in the trade pool being aggregated: the bound is only as good as S0 holds across the events being summed.

S0 is not diagnosable from trade data alone. Of the theorem's load-bearing assumptions, S2 is checkable via the benchmark function, and the genuine-exchange conditions (i)–(iv) of Definition 2 are checkable via trade records. S0 is the one assumption that cannot in principle be verified from the trade pool because it concerns the relationship between subjective valuation and realized exercise, neither of which is directly observable in the sacrifice channel. No structural mechanism in this paper or its companion [Lasser, 2026d, Need Sufficiency] diagnoses S0 failure; calibration of S0 is the work of institutional attestation (Remark 18) acting on data sources outside the sacrifice channel itself (e.g., longitudinal exercise studies, market-segment behavioural analysis, controlled-relaxation testing of the calibration chain). Open Question 7 is the formal extension; the present limitation is that the bound's quality is bounded by the institutional-attestation layer's quality on S0.

Remark 3 (Common-scale convention for the calibration chain). *The chain of inequalities $\text{vol}_R^{[W]}(Y) \geq U_i(Y) \geq U_i(X) \geq \Delta \text{vol}_P(X)$ presupposes a common scale on which the four quantities are comparable. We adopt the poset-measure scale: vol_P is the framework's measure-theoretic primitive ([Lasser, 2026f, Proposition 1]), and vol_R inherits its unit from vol_P on the exercised sub-poset (Definition 3). S0 is the assumption that each agent's subjective valuation U_i is reported in this scale (i.e., $U_i(Y)$ has the same units as $\text{vol}_R^{[W]}(Y)$); S2 is the analogous as-*

sumption on the benchmarked side ($U_i(X)$ in the same units as $\Delta \text{vol}_P(X)$). Money price p enters through the money-sacrifice channel (Definition 7) only via $\Delta \text{vol}_P(X(p))$, which converts price to poset-measure units through the benchmark calibration of [Lasser, 2026f, Definition 2]; analogously, the time-sacrifice channel (Definition 8) converts the wage-valued quantity $r_i \cdot h$ to poset-measure units through $\Delta \text{vol}_P(r_i \cdot h)$. Currency and wage-time are inputs to the benchmark function whose poset-measure-valued output is what enters the calibration chain.

Definition 1 (Aggregate Trade Window). An aggregate trade window $[t_0, t_0 + T]$ is an observation period over which revealed-sacrifice events are collected. The window parameters are:

- T : window duration;
- N : number of observed sacrifice events in the window;
- $\{(i_n, X_n, Y_n, t_n)\}_{n=1}^N$: the event sequence.

The aggregate trade window plays a role analogous to the observation windows used elsewhere in the sequence [Lasser, 2026g, SCM]: it defines the timescale over which the framework accumulates sacrifice evidence.

2.2 Sacrifice events and the cancellation lemma

Definition 2 (Revealed-Sacrifice Event). A revealed-sacrifice event is a tuple (i, X, Y, t) where:

- i is the agent performing the sacrifice;
- X is a benchmarked capability (or capability bundle) with known $\Delta \text{vol}_P(X) \geq 0$, surrendered by agent i at time t ;
- $Y = \{Y_1, \dots, Y_k\}$ is an unbenchmarkd capability bundle acquired by agent i , satisfying the genuine-exchange conditions: (i) $X \cap Y = \emptyset$ (the agent does not surrender and reacquire the same capability); (ii) $Y \cap (H_i \setminus X) = \emptyset$ (the acquired bundle is novel relative to the agent's retained holdings $H_i \setminus X$, where H_i is the agent's pre-trade holdings); (iii) post-trade cooperative closure: no cooperative capability in the poset has participants spanning both Y and $H_i \setminus X$ (so post-trade, no new cross-bundle cooperative capability is activated by the combination); and (iv) pre-trade cooperative closure: no cooperative capability in the poset has participants spanning both X and $H_i \setminus X$ (so pre-trade, no cooperative capability straddles the surrendered and retained capability sets—only cooperatives contained entirely in X or entirely in $H_i \setminus X$ exist);
- t is the event timestamp.

Conditions (i) and (ii) characterize genuine trades: the agent exchanges distinct capabilities and acquires something novel. Conditions (iii) and (iv) together ensure that the exchange-to-bundle cancellation in the revealed-preference derivation is exact: (iii) rules out post-trade cross-bundle cooperatives that would inflate $U_i((H \setminus X) \cup Y)$ beyond $U_i(H \setminus X) + U_i(Y)$; (iv) rules out pre-trade cross-bundle cooperatives that would inflate $U_i(H)$ beyond $U_i(H \setminus X) + U_i(X)$. Both closures are needed symmetrically for the cancellation of Lemma 1 to hold. Trades violating (i) or (ii) are degenerate; trades violating (iii) or (iv) carry revealed-preference information about the marginal contribution of Y in context rather than $\text{vol}_R(Y)$ in isolation, but the theorem's isolated- $\text{vol}_R(Y)$ interpretation and Lemma 1 require all four conditions. Contextual-quantity salvage is outside this paper's formal scope and is flagged as a future-work direction in Section 8. The event emits the signal:

$$\text{vol}_R^{\text{lower}}(Y) \geq \Delta \text{vol}_P(X)$$

under assumptions S0 (calibration), S1 (free choice), S2 (benchmark grounding), and S4(a) (valuation additivity) of Section 2.1. S4(a) is needed for the exchange-to-bundle cancellation (Lemma 1).

Lemma 1 (Exchange-to-Bundle Cancellation). Let H be a finite capability holding set with $X \subseteq H$, and let Y be a finite capability set satisfying the genuine-exchange conditions (i)–(iv) of Definition 2 relative to H : (i) $X \cap Y = \emptyset$; (ii) $Y \cap (H \setminus X) = \emptyset$; (iii) no cooperative capability in the poset has participants spanning both Y and $H \setminus X$; (iv) no cooperative capability in the poset has participants spanning both X and $H \setminus X$. Under S4(a) (global valuation additivity: $U_i(S) = \sum_{c \in S} u_i(c)$ for any

finite $S \subseteq P$, with cooperative capabilities included as explicit nodes in S when their participants are all in S):

$$U_i((H \setminus X) \cup Y) - U_i(H) = U_i(Y) - U_i(X).$$

Equivalently, $U_i((H \setminus X) \cup Y) \geq U_i(H) \iff U_i(Y) \geq U_i(X)$.

Proof. By S4(a), $U_i(A) = \sum_{c \in A} u_i(c)$ for any finite capability set A . To apply this cleanly we must track cooperative capabilities carefully: under S4(a), a cooperative capability c_{coop} contributes $u_i(c_{\text{coop}})$ to $U_i(A)$ whenever $c_{\text{coop}} \in A$, which holds iff all of c_{coop} 's participants are in A .

Pre-trade state H . $H = (H \setminus X) \sqcup X$. By (iv), no cooperative capability has participants spanning X and $H \setminus X$, so every cooperative capability in H has participants entirely in X or entirely in $H \setminus X$. Hence

$$U_i(H) = U_i(H \setminus X) + U_i(X),$$

where each term includes cooperatives formable within that set alone.

Post-trade state $(H \setminus X) \cup Y$. By (ii) this is a disjoint union: $(H \setminus X) \sqcup Y$. By (iii), no cooperative capability has participants spanning Y and $H \setminus X$, so every cooperative in the post-trade state has participants entirely in Y or entirely in $H \setminus X$. Hence

$$U_i((H \setminus X) \cup Y) = U_i(H \setminus X) + U_i(Y).$$

Cancellation. Subtracting, the $U_i(H \setminus X)$ terms cancel, leaving $U_i((H \setminus X) \cup Y) - U_i(H) = U_i(Y) - U_i(X)$.

Condition (i) makes $U_i(Y)$ and $U_i(X)$ refer to distinct capability sets; violating (i) would cause $X \cap Y$ terms to cancel trivially on both sides, yielding no information. ■

Remark 4 (Cooperative closure of capability sets). *Throughout this section, “ $U_i(S)$ ” for a finite capability set S is computed via S4(a) over the cooperative-closure of S : the set S^* obtained by adding to S every cooperative capability $z \in P$ all of whose participants are already in S . Concretely, $U_i(S) = \sum_{c \in S^*} u_i(c)$, where the sum runs over the closed set rather than the bare S . Cooperatives whose participants are not all in S are not added, even if some are. The genuine-exchange conditions (iii) and (iv) of Definition 2 ensure cooperative closures of $H \setminus X$, X , Y , H , and $(H \setminus X) \cup Y$ behave compatibly: condition (iv) makes $H^* = (H \setminus X)^* \sqcup X^*$ (no cooperative spans the $X/H \setminus X$ split), and condition (iii) makes $((H \setminus X) \cup Y)^* = (H \setminus X)^* \sqcup Y^*$ (no cooperative spans the $Y/H \setminus X$ split). The cancellation of Lemma 1 runs over these closed sets.*

Remark 5 (Bundle-for-bundle trades and mixed-polarity events). *Definition 2 permits X to be a benchmarked capability or bundle, and Y is always an unbenchmarkd bundle, so bundle-for-bundle trades are formally supported: the genuine-exchange conditions (i)–(iv) and Lemma 1 operate on set-level X and Y regardless of whether either side is singleton. When X is a bundle, $\Delta \text{vol}_P(X)$ is the benchmark function evaluated on the bundle as a single entity (typically via a transaction price or composite benchmark covering the bundle), inheriting whatever internal cooperative structure X carries by the cooperative-closure conditions.*

A complication this paper does not formally resolve is the mixed-polarity bundle-for-bundle case in which the same trade carries need-side and want-side polarities on different components of the surrendered bundle. A canonical example: an agent who moves from one home to another surrenders a bundle of capabilities (location, neighborhood, school district, commute, etc.) and acquires a different bundle of the same capability dimensions. Some dimensions are need-side (retention without shelter is degrading on the shelter dimension), others are want-side (the new home's kitchen is preferred but retention was non-degrading on that dimension). Under the current architecture, $S1$ admits such a trade whenever any non-degrading alternative existed (retention via refinancing, renting elsewhere, etc.), and the full $\Delta \text{vol}_P(X)$ enters $\text{vol}_R^{\text{lower}}$ as want-side preference revelation. This is conservative for $\text{vol}_R^{\text{lower}}$ (it cannot inflate the lower bound) but loose for the companion paper's need-access cost diagnostic [Lasser, 2026d, Need Sufficiency], which counts the entire $\Delta \text{vol}_P(X)$ of a mixed event as need-access cost and so overstates true need-access cost when most of the bundle is want-side.

The natural follow-up is an event-local bundle decomposition that splits $\Delta \text{vol}_P(X)$ across the surrendered bundle's component capabilities with polarity tags, yielding a need-share and a want-share separately. This extension is flagged as an open direction (Section 8) and is the operational form

of the side-move-versus-upgrade disambiguation problem (a trade that is a side-move on most dimensions and an upgrade on one is currently indistinguishable on the ledger from a trade that is an upgrade on most dimensions and a side-move on one).

Justification (revealed-preference inequality). The transfer from the observed trade to the framework-level bound $\text{vol}_R^{\text{lower}}(Y) \geq \Delta \text{vol}_P(X)$ proceeds in three steps.

1. **Exchange preference (revealed-preference observation).** When agent i trades X for Y , revealed-preference theory [Samuelson, 1938] establishes that the post-trade state is weakly preferred to the pre-trade state: $U_i((H \setminus X) \cup Y) \geq U_i(H)$, where H is the agent’s pre-trade holdings.
2. **Reduction to bundle-level comparison.** By Lemma 1 under S4(a) and the genuine-exchange conditions of Definition 2, the exchange comparison reduces to $U_i(Y) \geq U_i(X)$.
3. **Calibration chain.** Given $U_i(Y) \geq U_i(X)$, the bridge to the framework-level bound uses:
 - Sacrifice-side grounding (S2): $U_i(X) \geq \Delta \text{vol}_P(X)$, which holds by construction when vol_P is the operational target on the benchmarked subspace.
 - Acquisition-side calibration (S0): $U_i(Y) \leq \text{vol}_R^{[W]}(Y)$, asserting that agents do not systematically overvalue unbenchmarked bundles relative to their *window-active* exercise contribution. The window-active form is what the observation channel actually bounds: an acquisition at t_n carries no instantaneous exercise witness, but the bound is still valid on the cumulative window quantity $\text{vol}_R^{[W]}$ whenever the agent uses Y during W .

Chaining: $\text{vol}_R^{[W]}(Y) \geq U_i(Y) \geq U_i(X) \geq \Delta \text{vol}_P(X)$.

Remark 6 (S4(a) is a genuine assumption, not a consequence). *S4(a) is a global additive-separability assumption on U_i . It is stronger than the rationality-only premises of Samuelson’s original revealed-preference theorem [Samuelson, 1938], which establishes preferences over states rather than additivity over bundles. In applied microeconomics [Mas-Colell et al., 1995], bundle-level inference of the kind the sacrifice channel performs typically invokes either additive separability or quasilinearity in a numeraire. The paper uses a special case implicitly via the time-sacrifice channel (Definition 8), which values time at wage rate r_i —quasilinearity in wage-scaled time, a form of valuation additivity. S4(a) is the explicit assumption that makes Lemma 1 go through; it is not a consequence of S0–S3.*

Complementarities that would break pure additivity in general consumer theory are absorbed into the poset as cooperative nodes via M6 [Lasser, 2026f, Proposition 1]: cooperative capabilities are themselves capabilities with their own weights, so bundle-specific cooperative contributions appear as separate summands in U_i . S4(a)’s additivity is compatible with this structure because cooperative terms are explicit capabilities, not hidden non-linearities.

The cooperative-absorption argument is framework-internal, not a general claim about real-world valuation functions. *S4(a) is plausible in this paper because the GFM framework’s poset construction is disciplined to internalize complementarities as cooperative capabilities (M6); it is not the general claim that any agent’s valuation decomposes additively over bundle elements. Outside the framework, S4(a) would be a strong empirical hypothesis about real consumers; inside the framework, its plausibility is inherited from the construction discipline that puts each empirically-observed complementarity into the poset explicitly as a cooperative node. A poset that fails to do this (complementarities present in the agent’s valuation but absent from the poset’s cooperative structure) violates S4(a) by construction; the assumption is therefore as strong as the poset-construction discipline allows it to be, and no stronger. This is the architectural reason S4(a) appears as an explicit named assumption rather than a derived consequence, and the reason its operational adequacy is a property of the poset-construction governance layer, not of agent behaviour per se.*

2.3 Privacy properties

The channel’s privacy properties split cleanly into two categories: *formal* properties that follow directly from the observation-model construction (stated below as Theorem 1), and *institutional*

preconditions that require deployment-level infrastructure to hold operationally (stated as Proposition 1). We state each category separately so that the theorem-level content covers only mathematical consequences of the construction, while institutional dependencies are flagged as preconditions rather than proved facts.

Theorem 1 (Formal Privacy Properties of Revealed-Sacrifice Observation). *The revealed-sacrifice channel satisfies the following three formal privacy properties, each following from the observation tuple (Definition 2) together with the commitment protocol of [Lasser, 2026a, Definition 6]:*

- (P1) **No interior access.** *The framework never observes what the agent values privately, only what they commit to publicly through sacrifice. The observation channel records the tuple (i, X, Y, t) , not the agent’s internal valuation function U_i .*
- (P3) **Discretization as a feature.** *Trade events are sparse by construction. The channel’s emission rate is bounded by trade frequency, not by observation intensity: each sacrifice event produces one output tuple, and no continuous stream is emitted between events. (This is an emission-rate bound, not a Shannon-information-rate bound; the per-event payload tuple (i, X, Y, t) has finite information content under three standing assumptions: the capability poset P and category alphabet $\mathcal{C}$ are finite at evaluation time, the agent universe is finite, and timestamps are recorded at a public bounded precision (i.e., t takes values on a finite discretization grid spanning the trade window). Under all three, “finite” is bounded by the per-event payload size; absent any one—e.g., real-valued timestamps with unbounded precision—the per-event payload is not bounded uniformly.)*
- (P4) **Commitment-layer composability.** *The commitment protocol of [Lasser, 2026a, Definition 6] extends to revealed-sacrifice events: a sacrifice event can be committed as a proof-of-trade exposing only the vol_R -category and Δvol_P -magnitude, with counterparty and the specific benchmarked/unbenchmarkd goods hidden by the commitment’s hiding property (Section 6). Note that $\Delta\text{vol}_P(X)$ is a public function of the surrendered purchasing power on the money channel, so committing $\Delta\text{vol}_P(X)$ is informative about the price p via benchmark inversion—the commitment hides counterparty and specific goods, not price. Stronger price-hiding requires the hiding-variant rollup of Proposition 5, which releases only an aggregate threshold rather than per-event Δvol_P .*

Proof. Each property is established from the structure of the observation channel (Definition 2) and the aggregate bound (Theorem 2).

(P1) *No interior access.* The channel’s output is, by Definition 2, the tuple (i, X, Y, t) . The agent’s internal valuation U_i appears only as a bridge quantity in the calibration chain ($S_0 + S_1 + S_2 + S_4(a)$); it is never recorded, transmitted, or stored by the framework. The aggregate bound (Theorem 2) depends only on $\{(\Delta\text{vol}_P(X_n), Y_n)\}$, so the framework’s behaviour is identical under any two agent-valuation functions producing the same trade sequence. No interior access is required or obtained.

(P3) *Discretization as a feature.* The channel emits at most one output tuple per sacrifice event and produces no continuous stream otherwise. Over a window of duration T the total number of channel emissions is exactly the number of observed events, so the channel’s *emission rate* is $|\text{events}|/T$. Unlike continuous observation (which scales with sampling rate), the emission rate is bounded above by the agents’ trade frequency (a behavioural rather than observational parameter), so increasing observation intensity does not increase the emission rate. (Each emission carries finite information about the tuple (i, X, Y, t) ; this is an emission-rate claim, not a Shannon-information-rate claim.)

(P4) *Commitment-layer composability.* Each sacrifice event admits a commitment $\Gamma = \text{Commit}((\Delta\text{vol}_P(X), \text{category}(Y)), r)$ using the protocol of [Lasser, 2026a, Definition 6], accompanied by a zero-knowledge proof [Lasser, 2026a, Definition 7] that Γ binds the event-local objectively verifiable facts: non-negativity of $\Delta\text{vol}_P(X_n)$, bundle containment $Y_n \subseteq C_{\text{category}(Y_n)}$ (Definition 9(d)), and a verifiable link to an underlying trade record. The proof does not certify “valid revealed-sacrifice event” in the strict sense (all of $S_0, S_1, S_2, S_3, S_4(a)$, and the genuine-exchange conditions, with $S_4(b)$ and S_5 additionally for Part B); the non-objective hypotheses remain external attestations (Remark 17). This is the event-level certification; $S_4(b)$ is not an event-level property but a batch-level structural fact about the category partition, established via pairwise-distinct category labels in Proposition 5 and inherited from the poset-independence of distinct partition components

(Definition 10(i)). S_0 , S_1 , S_3 , $S_4(a)$, and S_5 are *not* ZK-verifiable under the default commitment (Remark 17); S_3 in particular requires either external attestation that the bundle’s hedonic decomposition is non-degenerate, or a richer commitment exposing decomposition fields. Proposition 4 establishes that the Part A aggregate bound is computable from committed events with *pairwise-distinct* category labels, together with the category-partition structure; batches with colliding labels require the operational responses of Remark 19 (sub-batch partitioning, finer public partition, or richer commitments). The committed form reveals only the sacrifice magnitude and bundle category; agent identity, specific X and Y , and counterparty remain hidden by the hiding property of Commit [Pedersen, 1991, Lasser, 2026a]. ■

Proposition 1 (Institutional Preconditions for Revealed-Sacrifice Observation). *Two additional privacy properties hold operationally when the deployment environment supplies the corresponding institutional infrastructure. These are not formal consequences of the channel’s construction; they are preconditions the framework relies on but does not itself construct.*

(P2) Consent via participation (institutional). *The aggregate lower bound sums event-magnitudes (Theorem 2, Part A) or per-capability max-attribution quantities (Part B) over observed events; in both Parts an agent with zero observed sacrifice events contributes zero to the aggregate. This zero-contribution property is formal and holds under Part A’s lighter hypothesis stack as well as Part B’s; we state P2 in its Part-A form to keep the privacy claim independent of Part B’s stronger assumptions. The operational interpretation as “consent” additionally requires that participation in markets and labour exchanges be legally and socially opt-in; in settings where participation is coerced, the formal zero-contribution property holds but the consent interpretation does not.*

(P5) Third-party observability (institutional). *Money-sacrifice events are observable from public-facing ledger data (market exchanges, payment rails, public commitments) with no agent participation required, assuming such public-ledger infrastructure exists. The price p is directly readable from public ledger data; the sacrifice magnitude $\Delta\text{vol}_P(X)$ is then computed from p via the benchmark function on the surrendered purchasing power (Definition 7). “Directly readable” refers to p ; Δvol_P is a function of p through a public benchmark, not a separately-disclosed quantity.*

Time-sacrifice events are observable from public-ledger data only under an additional precondition: the agent’s outside-option wage rate r_i (Definition 8) must be computable from public inputs—e.g., a public wage schedule, occupational census with public imputation rules, or the agent’s own published rate. Where r_i is private, the hours h may be observable but the sacrifice magnitude $\Delta\text{vol}_P(X(r_i \cdot h))$ is not third-party computable; P5 then holds for the money channel and for public-wage-imputation time events, but not for fully private time-sacrifice valuations.

Where public-ledger infrastructure does not exist at all, P5 fails operationally on both channels even though the construction is unchanged. This contrasts with controlled relaxation [Lasser, 2026h, Controlled Relaxation] (framework must impose a test) and with commitment-protocol verification [Lasser, 2026a, Exogenous Verification] (agent must generate a proof). P5 does not extend to private time-sacrifice activities (parenting, contemplation), which require voluntary disclosure regardless of ledger infrastructure and regardless of wage imputation.

Remark 7 (Why the split). *Theorem 1 restricts itself to properties that are mathematical consequences of the channel’s construction. Proposition 1 catalogues the deployment conditions under which P2 and P5 hold operationally. Treating P2 and P5 as institutional preconditions rather than as theorem consequences is the honest framing: they are load-bearing for how the channel behaves in practice, but they depend on facts about markets, legal systems, and public infrastructure that lie outside the paper’s formal scope.*

3 Aggregate Lower Bound Theorem

3.1 Realized capability volume (vol_R)

Before stating the aggregate theorem, we make the object being bounded precise. vol_R is the poset measure evaluated on the *exercised sub-poset*: the restriction of the capability space to capabilities that are currently being exercised.

Definition 3 (Exercised Sub-Poset and Realized Capability Volume). *Let P be the full capability poset with the poset measure vol_P [Lasser, 2026f, Definition 7] satisfying axioms M1–M6 [Lasser, 2026f, Proposition 1].*

Instantaneous form. *For each time t , the exercised sub-poset $P_t^{\text{ex}} \subseteq P$ is the set of capabilities being exercised at time t —a latent structural fact about the system, independent of any observation mechanism:*

$$P_t^{\text{ex}} = \{d \in P : d \text{ is exercised at time } t\},$$

where “exercised at t ” is formalized semantically via the counterfactual exercise condition: d is exercised at t iff some realized cooperative output in the lookback window $[t - \Delta, t]$ becomes un-realizable when d is removed from P . This is the latent structural notion; it does not depend on any perimeter or measurement mechanism. Appendix A (Definition 11) gives the operational realization of this counterfactual within an internal observation perimeter $O \subseteq P$, where the indicator is computable; outside O the counterfactual remains the latent definition of P_t^{ex} but is not directly measured by the indicator. The lookback width Δ and the indicator’s other parameters are specified in Definition 11; “at t ” here denotes that t is the right endpoint of the lookback, not that the exercise happens instantaneously at t . The instantaneous realized capability volume is the poset measure restricted to this sub-poset:

$$\text{vol}_{R,t}(S) = \text{vol}_P(S \cap P_t^{\text{ex}}) \quad \text{for any } S \subseteq P.$$

Window-active form (union over time). *For a trade window $W = [t_0, t_0 + T]$ (Definition 1), the window-active exercised sub-poset is the union of the instantaneous exercised sub-posets across the window,*

$$P_W^{\text{ex}} = \bigcup_{t \in W} P_t^{\text{ex}},$$

and the corresponding window-active realized capability volume is

$$\text{vol}_R^{[W]}(S) = \text{vol}_P(S \cap P_W^{\text{ex}}) \quad \text{for any } S \subseteq P.$$

$\text{vol}_R^{[W]}$ counts a capability as contributing to realized volume if it was exercised at any time during W ; the instantaneous form $\text{vol}_{R,t}$ counts only capabilities exercised exactly at t . Both inherit the measure structure of vol_P restricted to the corresponding sub-poset, and the axiom-analogue analysis of Section 5 applies to each.

Convention. Unless otherwise noted, vol_R without a time subscript refers to the window-active form $\text{vol}_R^{[W]}$ for the ambient trade window W . This is the quantity the sacrifice channel (Theorem 2) actually bounds: a sacrifice event at t_n records an acquisition, not an instantaneous exercise witness, so the theorem’s lower bound is stated against $\text{vol}_R^{[W]}$ —which captures any exercise of Y_n anywhere in W —rather than against the instantaneous $\text{vol}_{R,t}$ at a particular evaluation time. Assumption S0 (Section 2.1) is correspondingly stated directly in window-active form: $U_{i_n}(Y_n) \leq \text{vol}_R^{[W]}(Y_n)$.

Remark 8 (vol_R is latent; $\text{vol}_R^{\text{lower}}$ is the bound). vol_R (in either form) is defined structurally as a ground-truth quantity about the system; computing it directly requires evaluating the counterfactual exercise condition for every capability at every relevant time, which the Introduction rejects as intractable and privacy-incompatible at scale. The revealed-sacrifice observation channel does not observe P_t^{ex} or P_W^{ex} directly; it produces a lower bound $\text{vol}_R^{\text{lower}}$ on the window-active $\text{vol}_R^{[W]}$ restricted to an observed sub-space $U \subseteq P$. Inside the observation perimeter O of Appendix A, direct evaluation of $e_t(d)$ is permitted (no privacy barrier within consented operational scope) and contributes to a combined diagnostic $\text{vol}_R^{\text{combined}} = \text{vol}_R^{\text{exact}}(O) + \text{vol}_R^{\text{lower}}(P \setminus O)$,

which is a valid lower bound on $\text{vol}_R(P)$ when O is boundary-closed from $P \setminus O$ in the full poset-independence sense (neither cooperative-composition nor subsumption links cross the boundary; see Definition 11 conditions BC1/BC2). Outside that regime, the compositional lower-bound interpretation of $\text{vol}_R^{\text{combined}}$ is not established by this paper and the quantity should be treated as a heuristic composite (Appendix A, Remark 23). vol_R itself is the latent target; $\text{vol}_R^{\text{lower}}$ (and $\text{vol}_R^{\text{exact}}$ on O) are the observation-channel outputs the framework actually computes.

3.2 Event-local decomposition

Definition 4 (Event-Local Bundle Decomposition). *For each revealed-sacrifice event (i_n, X_n, Y_n, t_n) whose bundle Y_n is within-bundle poset-independent (no subsumption relations and no cooperative-composition links among the capabilities in Y_n ; see Definition 10(i) applied internally to Y_n), define the event-local decomposition coefficients $\alpha_{n,c} \geq 0$ for each capability $c \in Y_n$, satisfying:*

$$\sum_{c \in Y_n} \alpha_{n,c} = 1 \quad (\text{partition of unity within the bundle}).$$

The coefficients represent the share of bundle Y_n attributable to capability c , determined at the time of event n and fixed thereafter. Each event carries its own decomposition, independent of other events. The decomposition can be obtained from prior hedonic regression [Rosen, 1974] on similar trades, domain knowledge, or the equal-weight default $\alpha_{n,c} = 1/|Y_n|$.

Within-bundle poset-independence is the hypothesis that makes the partition of unity compatible with S5's share inequality. Under within-bundle independence, no cooperative node activates from within Y_n , so $\text{vol}_R(Y_n) = \sum_{c \in Y_n} \text{vol}_R(c)$ by axiom M4, and S5's $\alpha_{n,c} \leq \text{vol}_R(c)/\text{vol}_R(Y_n)$ sums to at most one across $c \in Y_n$ with equality achievable. Bundles with internal cooperative activation fall back to Part A only (see Remark 9).

Remark 9 (Why within-bundle independence is a Part B hypothesis). *If Y_n contains capabilities a, b whose cooperative composition z has all participants in Y_n , then S4(a) (Section 2.1) treats z as an explicit bundle summand, so $\text{vol}_R(Y_n) = \text{vol}_R(a) + \text{vol}_R(b) + \text{vol}_R(z) > \sum_{c \in Y_n} \text{vol}_R(c)$. S5's share inequality $\alpha_{n,c} \leq \text{vol}_R(c)/\text{vol}_R(Y_n)$ is then forced below $\text{vol}_R(c)/(\text{vol}_R(a) + \text{vol}_R(b) + \text{vol}_R(z))$ for each atomic $c \in Y_n$, and $\sum_{c \in Y_n} \alpha_{n,c} \leq [\text{vol}_R(a) + \text{vol}_R(b)]/\text{vol}_R(Y_n) < 1$, contradicting the partition of unity. Part B's per-capability attribution therefore requires within-bundle poset-independence. Part A places no such constraint: its bundle-level bound $\text{vol}_R(Y_n) \geq \Delta \text{vol}_P(X_n)$ is valid regardless of internal cooperative structure.*

Remark 10 (Event-local versus global regression). *Classical hedonic regression [Rosen, 1974] fits a global model across all observed trades simultaneously, producing the tightest decomposition but breaking the monotonicity of the max-attribution bound (Proposition 2) when new observations refit all coefficients. The event-local formulation fixes coefficients at observation time, sacrificing fit quality for the monotonicity guarantee.*

3.3 Central result

Theorem 2 (Aggregate B-to-C Lower Bound). *Fix a trade window $W = [t_0, t_0 + T]$ (Definition 1) and a sequence of revealed-sacrifice events $\{(i_n, X_n, Y_n, t_n)\}_{n=1}^N$ with $t_n \in W$, whose bundles jointly cover a subset U of unbenchmarked capability-space. All vol_R quantities below refer to the window-active form $\text{vol}_R^{[W]}$ (Definition 3): the theorem bounds the realized volume accumulated during W , not the instantaneous vol_{R_t} at any single evaluation time.*

Part A (Main result — bundle-level lower bound; requires S0–S2 and S4(a)). *Each bundle Y_n is taken as cooperative-closed: Y_n contains a cooperative capability z iff all participants of z are in Y_n . For each event n :*

$$\text{vol}_R^{[W]}(Y_n) \geq \Delta \text{vol}_P(X_n).$$

For N events with pairwise poset-independent bundles (in the sense of Definition 10(i): no subsumption relation and no cooperative-composition link connects any capability in Y_n to any capability in $Y_{n'}$ for $n \neq n'$). We additionally require set-disjointness $Y_n \cap Y_{n'} = \emptyset$ for $n \neq n'$, which is implicit

when subsumption is taken as a strict partial order on distinct capabilities; we make it explicit here so that the proof’s M4 step does not have to invoke reflexivity of the relation:

$$\text{vol}_R^{U,[W]} \geq \sum_{n=1}^N \Delta \text{vol}_P(X_n),$$

where $U = \bigcup_n Y_n$ and $\text{vol}_R^{U,[W]}$ is the window-active realized capability volume restricted to U . Pairwise poset-independence makes U itself cooperative-closed (no cooperative spans any pair $Y_n, Y_{n'}$, so each cooperative whose participants lie in U has participants entirely in some single Y_n). Part A is the paper’s substantive result: the vol_P -to- vol_R lower bound per unit of observed sacrifice magnitude.

Part B (Conditional refinement — per-capability attribution; requires S0–S5, within-bundle poset-independence, and positive sacrifice magnitude). Restrict attention to events n with $\Delta \text{vol}_P(X_n) > 0$ whose bundles Y_n are within-bundle poset-independent (Definition 4). Events with $\Delta \text{vol}_P(X_n) = 0$ carry no Part-B information and are excluded from the attribution. When bundles overlap and per-capability attribution is desired, define for each capability $c \in U$:

$$\text{vol}_R^{\text{lower}}(c) = \max_{n: c \in Y_n} \alpha_{n,c} \cdot \Delta \text{vol}_P(X_n).$$

Then, provided the capabilities $c \in U$ are pairwise poset-independent:

$$\text{vol}_R^{U,[W]} \geq \sum_{c \in U} \text{vol}_R^{\text{lower}}(c).$$

Part B is conditional on S5—a direct share inequality on the quantity being bounded (see Remark 1)—and should be read as a refinement when admissible, not as a result comparable in strength to Part A.

Proof. Part A. Fix an event n with $t_n \in W$. By S1, agent i_n chose Y_n over X_n while both were available. Revealed preference gives $U_{i_n}((H_n \setminus X_n) \cup Y_n) \geq U_{i_n}(H_n)$ for pre-trade holdings H_n . By Lemma 1 under S4(a) and the genuine-exchange conditions of Definition 2, this reduces to $U_{i_n}(Y_n) \geq U_{i_n}(X_n)$. By S0 (valuation-bounded-by-exercise, stated in window-active form), $U_{i_n}(Y_n) \leq \text{vol}_R^{[W]}(Y_n)$. By S2 (benchmark grounding), $U_{i_n}(X_n) \geq \Delta \text{vol}_P(X_n)$. Chaining:

$$\text{vol}_R^{[W]}(Y_n) \geq U_{i_n}(Y_n) \geq U_{i_n}(X_n) \geq \Delta \text{vol}_P(X_n),$$

which is the per-event bound. The window-active form is essential: the sacrifice channel observes acquisition of Y_n at t_n , not exercise at any particular instant. S0’s consequent is therefore stated as a bound on $\text{vol}_R^{[W]}(Y_n)$ —the cumulative exercise witness for Y_n across W —not on the instantaneous $\text{vol}_{Rt}(Y_n)$ at any single evaluation time t , which could be zero even when Y_n was exercised earlier or later in W .

Suppose the N bundles $\{Y_n\}$ are pairwise poset-independent and each is cooperative-closed (no cooperative is split between Y_n and any other bundle, and each Y_n contains its own internal cooperatives). Then the sub-posets carried by the bundles are disconnected components of the sub-poset on $U = \bigcup_n Y_n$, and U inherits cooperative closure from the per-bundle closure plus pairwise independence. Axiom M4 of [Lasser, 2026f, Proposition 1] makes the poset measure additive over poset-disjoint cooperative-closed components; $\text{vol}_R^{[W]}$ inherits M4 (Section 5, Theorem 3), so $\text{vol}_R^{U,[W]} = \sum_n \text{vol}_R^{[W]}(Y_n)$. Summing the per-event bounds:

$$\text{vol}_R^{U,[W]} = \sum_{n=1}^N \text{vol}_R^{[W]}(Y_n) \geq \sum_{n=1}^N \Delta \text{vol}_P(X_n),$$

establishing Part A.

Part B. Part B is stated for events with $\Delta \text{vol}_P(X_n) > 0$ and bundles Y_n that are within-bundle poset-independent (Definition 4); events with zero sacrifice carry no information, and bundles with internal cooperative activation fall back to Part A (Remark 9). Fix an admissible event n and its bundle Y_n . By Part A, $\text{vol}_R(Y_n) \geq \Delta \text{vol}_P(X_n) > 0$, so $\text{vol}_R(Y_n) > 0$ and division is well-defined.

S3 (bundle coherence) additionally ensures the hedonic decomposition of Y_n is non-degenerate, so $\alpha_{n,c}$ is well-specified.

Within-bundle poset-independence plus M4 (Section 5, Theorem 3) gives $\text{vol}_R(Y_n) = \sum_{c \in Y_n} \text{vol}_R(c)$: no cooperative node activates from within Y_n , so the bundle's vol_R decomposes atomically. S5 then asserts that the event-local coefficients are lower bounds on the true atomic shares:

$$\alpha_{n,c} \leq \text{vol}_R(c) / \text{vol}_R(Y_n), \quad c \in Y_n,$$

which is feasible with $\sum_{c \in Y_n} \alpha_{n,c} = 1$ precisely under within-bundle independence. Rearranging and applying the Part A per-event bound:

$$\text{vol}_R(c) \geq \alpha_{n,c} \cdot \text{vol}_R(Y_n) \geq \alpha_{n,c} \cdot \Delta \text{vol}_P(X_n),$$

for every $c \in Y_n$. If c appears in multiple admissible bundles, each quantity $\alpha_{n,c} \cdot \Delta \text{vol}_P(X_n)$ is a valid lower bound on $\text{vol}_R(c)$; taking the maximum preserves validity:

$$\text{vol}_R^{\text{lower}}(c) := \max_{n: c \in Y_n} \alpha_{n,c} \cdot \Delta \text{vol}_P(X_n) \leq \text{vol}_R(c).$$

(Using the max rather than a sum is what underwrites the monotonicity of Proposition 2: additional events can only weakly increase a max.)

By the Part B hypothesis, the capabilities $c \in U$ are pairwise poset-independent, so a second application of M4 (to the poset of capabilities, not bundles) gives global additivity:

$$\text{vol}_R^{U,[W]} = \sum_{c \in U} \text{vol}_R(c) \geq \sum_{c \in U} \text{vol}_R^{\text{lower}}(c).$$

■

Remark 11 (Why poset-independence, not set-disjointness). *Set-disjointness* ($Y_1 \cap Y_2 = \emptyset$) ensures no capability appears in two bundles, but does not rule out subsumption or cooperative links across bundles. If a cooperative capability c_{coop} depends on capabilities in both Y_1 and Y_2 , its vol_R contribution is not attributable to either bundle alone. *Poset-independence* rules this out: under poset-independence, the sub-posets are disconnected components and vol_P (hence vol_R) decomposes additively by axiom M4 [Lasser, 2026f, Proposition 1].

Definition 5 (Subset and Category-Level $\text{vol}_R^{\text{lower}}$). *Under the Part B hypotheses (S0–S5) of Theorem 2, the per-capability lower bound $\text{vol}_R^{\text{lower}}(c)$ extends to any subset $A \subseteq P$ whose covered capabilities are pairwise poset-independent:*

$$\text{vol}_R^{\text{lower}}(A) := \sum_{c \in A \cap U} \text{vol}_R^{\text{lower}}(c), \quad \text{when } A \cap U \text{ is pairwise poset-independent,}$$

where U is the observed subset covered by bundles. In this regime the sum is a valid lower bound on $\text{vol}_R(A)$, via Part B applied to the sub-family $A \cap U$. When $A \cap U$ has internal poset-dependencies, the same expression can be formed but is an accounting score, not a theorem-certified lower bound on $\text{vol}_R(A)$. In particular:

- For a bundle category $C_j \in \mathcal{C}$ (Definition 10), $C_j \cap U$ is pairwise poset-independent only when the public partition is refined below internal dependencies. For coarser partitions, $\text{vol}_R^{\text{lower}}(C_j)$ is an accounting score for the category.
- For any other capability subset A used by downstream diagnostics (restriction set, residual class, diagnostic cell), certified lower-bound status requires the same internal poset-independence of $A \cap U$.

Under Part A alone, $\text{vol}_R^{\text{lower}}$ is defined only at the per-bundle level; the global aggregate is $\text{vol}_R^{U,[W]} \geq \sum_n \Delta \text{vol}_P(X_n)$, and arbitrary subset-level $\text{vol}_R^{\text{lower}}(A)$ is not a Part A object. The companion paper's gap decomposition and wireheading diagnostic [Lasser, 2026d, Need Sufficiency] are stated under Part B semantics and report subset/cell sums as theorem-certified lower bounds only when the corresponding internal poset-independence condition is met; otherwise the sums are reported as diagnostic accounting scores.

3.4 B-to-C ratio and monotonicity

Definition 6 (B-to-C Ratio Under Revealed Sacrifice). *The B-to-C ratio under revealed-sacrifice observation is:*

$$\beta^{\text{lower}}(G, T) = \text{vol}_R^{\text{lower}}(G, T) / \text{vol}_P(G),$$

where G is a capability subset on which $\text{vol}_R^{\text{lower}}(G, T)$ is defined. The two reference choices are: (a) $G = U$, where U is the observed subset covered by admissible bundles (so $\text{vol}_R^{\text{lower}}(U, T)$ is the per-capability or sub-batch-supremum aggregate of Theorem 2); (b) $G = P$, the full capability-space denominator, with numerator $\text{vol}_R^{\text{lower}}(U, T)$ extended by zero outside U to yield a global conservative ratio. In both cases $\beta^{\text{lower}} \in [0, 1]$: the numerator is ≥ 0 by construction and is bounded above by $\text{vol}_P(G)$ because $\text{vol}_R^{\text{lower}}(G, T) \leq \text{vol}_R(G) \leq \text{vol}_P(G)$ (Part-A/Part-B inequalities chained with $\text{vol}_R \leq \text{vol}_P$). For arbitrary $G \subseteq P$ outside case (a) or (b), $\text{vol}_R^{\text{lower}}(G, T)$ requires the subset-extension hypotheses of Definition 5; without those, $\beta^{\text{lower}}(G, T)$ is an accounting score, not a theorem-certified ratio. T indexes the trade window.

Proposition 2 (Part B Monotone Accumulation Under Max-Attribution). *Under Part B hypotheses (S0–S5 and within-bundle poset-independence), define the per-capability lower bound as the maximum attribution across all observed events, with the zero-extension convention $\text{vol}_R^{\text{lower}}(c) = 0$ when c is in no observed bundle:*

$$\text{vol}_R^{\text{lower}}(c) = \max_{n: c \in Y_n} \alpha_{n,c} \cdot \Delta \text{vol}_P(X_n), \quad \text{or } 0 \text{ if no event covers } c.$$

Aggregating over a fixed common universe $U^* \supseteq \bigcup_{n \in E'} Y_n$ (any superset of the largest observed U across the event sets compared, with capabilities outside the observed bundles contributing zero by the convention above), $\text{vol}_R^{\text{lower}}(E) := \sum_{c \in U^*} \text{vol}_R^{\text{lower}}(c \mid E)$ satisfies: for event sets $E \subseteq E'$ both satisfying the Part B admissibility conditions,

$$\text{vol}_R^{\text{lower}}(E') \geq \text{vol}_R^{\text{lower}}(E).$$

Proof. Fix $c \in U$. Let $S_N(c) = \{n \leq N : c \in Y_n\}$ be the set of events in the first N observations that cover c . For $N' > N$, $S_{N'}(c) \supseteq S_N(c)$, and

$$\text{vol}_R^{\text{lower}}(N')(c) = \max_{n \in S_{N'}(c)} \alpha_{n,c} \Delta \text{vol}_P(X_n) \geq \max_{n \in S_N(c)} \alpha_{n,c} \Delta \text{vol}_P(X_n) = \text{vol}_R^{\text{lower}}(N)(c),$$

because a maximum over a superset is weakly larger. If $c \notin Y_n$ for any n then both sides are zero by convention and the inequality is trivial. Summing the per-capability inequalities over $c \in U$: $\text{vol}_R^{\text{lower}}(N') \geq \text{vol}_R^{\text{lower}}(N)$. ■

Proposition 3 (Part A Monotone Accumulation via Sub-Batch Supremum). *Under Part A hypotheses (S0–S2, S4(a), and the genuine-exchange conditions of Definition 2), define the Part A lower bound as the supremum over admissible pairwise-poset-independent sub-batches of the observed event set:*

$$\text{vol}_R^{\text{lower}^A}(E) := \sup_{B \in \mathcal{B}(E)} \sum_{n \in B} \Delta \text{vol}_P(X_n),$$

where $\mathcal{B}(E)$ is the family of sub-batches $B \subseteq E$ satisfying the Part A admissibility and pairwise poset-independence hypotheses of Theorem 2, with the supremum attaining 0 on the empty sub-batch. Then $\text{vol}_R^{\text{lower}^A}$ is non-decreasing in E : for event sets $E \subseteq E'$,

$$\text{vol}_R^{\text{lower}^A}(E') \geq \text{vol}_R^{\text{lower}^A}(E).$$

Proof. Any admissible sub-batch $B \subseteq E$ is also an admissible sub-batch $B \subseteq E'$ (admissibility is a property of B itself, not of its ambient event set). Hence the supremum over admissible sub-batches of E is a supremum over a subset of the admissible sub-batches of E' , and the supremum over a superset is weakly larger. ■

Remark 12 (Interpretation of Part A monotonicity). $\text{vol}_R^{\text{lower}^A}(E)$ is the tightest Part A lower bound the framework can extract from the event set E by choosing which sub-batch to apply Theorem 2

to. The supremum is over single sub-batches; computing it does not entail summing across multiple disjoint sub-batches of E , because doing so could double-count when an underlying capability appears in multiple sub-batches. Operationally, the framework selects the single sub-batch maximising the Part A sum; if multiple maximal admissible sub-batches partition the covered space into poset-disjoint components, their sum is also admissible (and equal to the supremum on that one combined sub-batch), but the principled definition is the supremum over sub-batches, not the sum across them. $\text{vol}_R^{\text{lower}^A}$ is what the paper refers to when it says “the constructed Part A lower bound is non-decreasing in observed admissible events.”

4 Two Sacrifice Channels: Money and Time

The sacrifice dimension splits into two observation channels. Money-sacrifice and time-sacrifice are not disjoint coverage spaces; they are *different modalities of the same underlying sacrifice substrate*, connected by cooperative-market exchange. The value of observing both is triangulation and robustness, with genuine strict inequality arising only at specific non-fungibility residuals.

4.1 Money-sacrifice channel

Definition 7 (Money-Sacrifice Event). A money-sacrifice event is a revealed-sacrifice event (i, X, Y, t) where the benchmarked sacrifice $X = X(p)$ is the purchasing-power capability bundle corresponding to surrendering price p : agent i pays p for an unbenchmarked bundle Y . The signal is:

$$\text{vol}_R^{\text{lower}}(Y) \geq \Delta \text{vol}_P(X(p)),$$

where $X(p)$ denotes the purchasing-power capability bundle of magnitude p and $\Delta \text{vol}_P(X(p))$ is the vol_P contribution of the surrendered bundle (computed via the public benchmark function on purchasing power).

Properties of the money channel:

- **Signal strength.** Price is a precise, quantitative signal. Market aggregation handles distributional loading (many agents trading the same category of goods produces a robust estimate).
- **Bias.** Privileges capabilities valued by wealthier agents. An agent with more purchasing power emits stronger signals per trade.
- **Third-party observability.** Money-sacrifice events leave public ledger traces (prices, timestamps, category labels); auditors can compute the *public-ledger flow aggregate* without agent cooperation (Property P5 of Proposition 1). Computing the *theorem-admissible* aggregate $\sum_n \Delta \text{vol}_P(X_n)$ that Theorem 2 certifies as a vol_R lower bound additionally requires the public S1-admissibility labelling and the other external-attestation outputs of Remark 18 (genuine-exchange conditions, S0 calibration, S4(a)). Ledger auditability is the data-availability property; admissibility-label auditability is a separate institutional property whose realization is governance-supplied.

4.2 Time-sacrifice channel

Definition 8 (Time-Sacrifice Event). A time-sacrifice event is a revealed-sacrifice event (i, X, Y, t) where the benchmarked sacrifice $X = X(r_i \cdot h)$ is the purchasing-power-equivalent bundle of foregone wage income $r_i \cdot h$, with r_i the agent’s market wage rate and h the foregone hours. Agent i foregoes h hours of wage-earning time to engage in an unbenchmarked activity (volunteering, parenting, craft practice, relationship-building, contemplation). The signal is:

$$\text{vol}_R^{\text{lower}}(Y) \geq \Delta \text{vol}_P(X(r_i \cdot h)),$$

where $X(r_i \cdot h)$ is the purchasing-power-equivalent bundle of magnitude $r_i \cdot h$ (so the time channel reduces to a money-channel evaluation through the wage-rate identification). Per-event admissibility ceiling. $\text{vol}_R^{[W]}(Y)$ is the window-active union (Definition 3), bounded above by $\sum_{c \in Y} w(c)$ regardless of how long Y is exercised. The calibration chain $\Delta \text{vol}_P(X(r_i \cdot h)) \leq U_i(X) \leq U_i(Y) \leq$

$\text{vol}_R^{[W]}(Y) \leq \sum_{c \in Y} w(c)$ (revealed preference plus S0 and S2) caps the event magnitude at the bundle's ceiling; long-duration time sacrifices that drive $r_i \cdot h$ above this ceiling indicate either S0 violation, hidden bundle content beyond the declared Y , or mis-imputed r_i , and the event is inadmissible. This is the central admissibility constraint of the time-sacrifice channel; treat it as part of the definition rather than a post-hoc remark (the cap is structural, not a filtering heuristic). See Remark 13 for the full discussion.

Remark 13 (S0 implicitly caps long-duration time sacrifices). $\Delta \text{vol}_P(X(r_i \cdot h))$ is monotone non-decreasing in h (under the standing assumption that the public benchmark function on purchasing power is monotone in price; this is a convention of the benchmark, not a separate theorem), while $\text{vol}_R^{[W]}(Y)$ is the window-active sum (Definition 3) bounded above by $\sum_{c \in Y} w(c)$ regardless of how long Y is exercised. A time-sacrifice event with $r_i \cdot h$ large enough that $\Delta \text{vol}_P(X(r_i \cdot h)) > \sum_{c \in Y} w(c)$ implies the agent is paying more (in benchmarked vol_P units) than the bundle's full window-active vol_R ceiling, which violates S0 ($U_i(Y) \leq \text{vol}_R^{[W]}(Y)$) for any reasonable identification $U_i(Y) \leq \sum_{c \in Y} w(c)$ (which holds when valuations are bounded by capability ceilings). Such overpayment events are rejected by S0 and excluded from the aggregate; the cap is automatic from the assumption stack rather than imposed by an explicit filter. Operationally, this means very-long-duration time sacrifices on small bundles indicate either (i) genuine S0 violation (the agent overvalues), (ii) hidden bundle content not in Y (the sacrifice is for a larger bundle than declared), or (iii) mis-imputed r_i . Each is a flag for review rather than a free pass to inflate $\text{vol}_R^{\text{lower}}$.

Properties of the time channel:

- **Signal strength.** Opportunity cost is a quantitative signal grounded in market wage rates. Every agent has a time endowment; agents in the labour market have a market-wage outside option, while non-market agents (next bullet) require an imputed or substitute valuation.
- **Bias.** More egalitarian than the money channel. Time is the scarcest capability every agent shares. A high-wage agent's time sacrifice has higher Δvol_P per hour, but every agent sacrifices time.
- **Non-market agents.** Agents not in the labour market (retirees, children, homemakers, the unemployed) have an undefined or zero market wage rate. The framework can use imputed wage rates [Aguiar and Hurst, 2007] or non-market time valuation (replacement cost) to extend coverage to these populations.
- **Public versus private sacrifice.** Publicly observable time-sacrifice (committed volunteer hours, time-stamped artefact production) is third-party observable at the data-availability level; computing the theorem-admissible time-channel aggregate also requires the public S1-admissibility labelling and the other external-attestation outputs of Remark 18, mirroring the money-channel split above. Private time-sacrifice (parenting, contemplation) is visible only via voluntary disclosure—the residual flagged in Property P5 of Proposition 1.

4.3 Substitutability via cooperative markets

Money and time are not disjoint sacrifice types. Cooperative markets exchange between them:

- **Time $\rightarrow$ money.** Employment converts labour hours to wage income at rate r_i . This is the definitional exchange rate already embedded in Definition 8.
- **Money $\rightarrow$ time.** Agents can hire others to invest labour hours (services, delegation) or purchase time-saving technology and infrastructure (automation, efficiency-enhancing tools). The exchange goes through the same cooperative market that sustains employment.

Because the two channels are linked by this exchange, an agent's total sacrifice budget can be measured in a common monetary unit via the wage rate, and the agent can shift sacrifice between modalities. A high-earning agent can substitute money for time by hiring help; a time-rich agent can substitute time for money by investing hours directly. Neither channel *dominates* the other—each is the natural view of the same underlying sacrifice from one side of the exchange.

The practical consequence: **the two channels triangulate the same capability space, they do not partition disjoint capability spaces.** Observing both yields:

- **Triangulation.** When money-sacrifice and time-sacrifice both indicate high $\text{vol}_R^{\text{lower}}$ on a bundle, the signals corroborate. Divergence between the channels is itself diagnostic: a capability with strong time-sacrifice evidence but no money-sacrifice evidence in a given window may indicate the capability is exercised by time-rich agents outside the monetary economy, or that market mechanisms have not formed around it.
- **Robustness.** Different agent populations emit different modal signals. Children, retirees, and non-market agents emit time signals with imputed wage rates; wage- constrained high earners emit money signals. Observing both modalities extends the bound across populations that either alone would miss.

4.4 Non-fungibility residuals

The substitutability in §4.3 is not total. Two informal observations identify capabilities where one channel produces sacrifice events that the other does not produce on the *same* capability. We label these as *examples* rather than propositions because the arguments rely on the intuitive notion “*c* generates an event of channel *X* on *c*” rather than on a formal predicate; the intent is to motivate that the joint aggregate strictly improves on either channel alone in non-trivial cases, not to prove a structural impossibility.

Example 1 (Agent-specific-time non-substitutability, informal). Take $c^* = \text{“sustained mentoring relationship with a specific individual.”}$ The capability is constituted by the interaction between agent and mentee; it is not transferable to a proxy. Money-channel events on c^* are not produced: purchasable access goods (introductions, scheduled mentor time, coaching infrastructure) are distinct bundles whose vol_R -content lies in those access bundles, not in the relational c^* itself. A time-sacrifice event of h hours allocated to the relationship at wage rate r_i (Definition 8) produces $\text{vol}_R^{\text{lower}}(Y) \geq \Delta \text{vol}_P(X(r_i \cdot h))$ with $c^* \in Y$, an event the money channel does not produce on c^* . The asymmetry is informal-existential: it identifies one capability where the time channel reaches c^* via a sacrifice event and the money channel does not, not a claim that all agent-specific-time capabilities evade the money channel.

Example 2 (Capital-concentrated non-substitutability, informal). Take $c^* = \text{“conduct a particle-physics experiment using a synchrotron.”}$ Exercise of c^* requires access to specialized infrastructure whose acquisition or lease cost is concentrated in a single transaction (institutional budget, grant award, facility time purchased). An agent accumulating time-sacrifice at wage rate r_i can only reach this capability by first converting time to money via employment and then purchasing access—a money-channel event on the access category. Direct time-sacrifice on c^* without capital access does not produce a $\text{vol}_R^{\text{lower}}$ signal on c^* . The asymmetry is informal-existential: it identifies one capability where the money channel reaches c^* via a sacrifice event and the time channel does not.

Remark 14 (Scope of the residual examples). Examples 1 and 2 are informal-existential—each exhibits one capability that motivates the claimed residual behaviour. They do not establish that the descriptive class “all agent-specific-time capabilities” or “all capital-concentrated capabilities” is uniformly channel-exclusive. Mixed-mode coverage of such capabilities can and does occur (a paid mentor also incurs time-sacrifice; a well-funded scientist can generate labor-attributed time events alongside capital-access events). The existential residuals are sufficient for Corollary 1’s strict-improvement claim, which is stated conditional on the trade window actually containing at least one residual-category event.

Corollary 1 (Joint Channel Triangulation). Let E_{money} and E_{time} denote the money-channel and time-channel event sets over a trade window W , and suppose the union $E = E_{\text{money}} \cup E_{\text{time}}$ satisfies the closure conditions of Theorem 2 Part A: the bundles $\{Y_n\}_{n \in E}$ are pairwise poset-independent across channels—no subsumption relation and no cooperative-composition link connects any capability in a money-channel bundle to any capability in a time-channel bundle (set-disjointness of bundles across channels is necessary but not sufficient).

Weak containment (under the stated closure hypotheses). The joint aggregate captures weakly more of U than either channel alone: any event observed through either channel alone is in the union, and the Part A lower bound under the sub-batch supremum is monotone in observed admissible events (Proposition 3). “Weak containment” here means weak monotonicity of the sub-batch supremum under event-set inclusion, conditional on the cross-channel pairwise poset-independence and Part A admissibility hypotheses above; it is not a claim of unconditional containment.

Strict improvement (conditional on observed coverage). *The non-fungibility residuals (Examples 1 and 2) make strict improvement possible: agent-specific-time capabilities lie in the coverage gap of the money channel alone, and capital-concentrated capabilities lie in the coverage gap of the time channel alone. Strictness over the money channel alone obtains when W contains at least one admissible time-only residual event with positive sacrifice magnitude ($\Delta \text{vol}_P(X_n) > 0$); strictness over the time channel alone obtains symmetrically with at least one admissible money-only residual event with positive magnitude. Zero-magnitude admissible events do not improve the sub-batch supremum and so do not produce strict improvement. The residuals establish the availability of strict improvement, not its automatic realization; whether any given window realizes strictness is a data-coverage question.*

Proof sketch. Weak containment under the stated closure hypothesis: any event observed through either channel alone is also observed through the union, and the cross-channel pairwise poset-independence ensures Part A’s aggregation is well-defined on E ; the Part A lower bound is monotone in observed admissible events under sub-batch supremum (Proposition 3), so the union’s bound is $\geq$ either single channel’s bound. Strict improvement on the residuals: Example 1 witnesses a capability captured by time but not by money; Example 2 witnesses a capability captured by money but not by time. Both residuals guarantee strict improvement of the joint aggregate over either channel alone when the trade window includes at least one event from the relevant residual category. Outside the residuals, strictness becomes a data-coverage statement: the trade window including admissible events in categories the other channel does not cover is *sufficient* for the joint aggregate to strictly exceed that channel alone (via the supremum monotonicity of Proposition 3); the converse direction is not claimed by this corollary, since strict improvement could in principle arise from other configurations of the sub-batch supremum that this proof does not enumerate. Substitutability in principle does not guarantee equality in practice; heterogeneous agent populations generically produce strict improvement, but a theoretical population using one channel exclusively could produce single-channel coverage equal to the joint aggregate. ■

Remark 15 (Cross-channel poset coupling). *The closure hypothesis rules out all three forms of cross-channel coupling: (a) a capability c appearing in both a money-channel bundle and a time-channel bundle (set-overlap), (b) subsumption across channels ($c \succeq c'$ with c in one channel’s bundle and c' in the other’s), and (c) cooperative composition linking capabilities across channels. When any of these occurs in practice, the operational responses of Remark 19 apply: partition events into maximal poset-independent sub-batches per channel and aggregate separately, or fall back to Part B’s per-capability max-attribution (which natively handles overlapping events via the max across events covering each capability, subject to Part B’s additional hypotheses).*

4.5 The residual intersection

What neither channel catches is the *residual class* (characterized in the companion paper [Lasser, 2026d, Need Sufficiency]): capabilities exercised purely privately, with no trade event, no publicly-observable time expenditure, and no sacrifice footprint of any kind. The residual class is the intersection of the coverage gaps of both channels after accounting for the cooperative-market substitutability of §4.3 and the non-fungibility residuals of Examples 1 and 2. This is a strictly smaller set than the union of the two channels’ individual coverage gaps: capabilities in one channel’s gap but reachable via the other (or via the substitution markets) are not residual.

5 Axiom Inheritance and the Non-Self-Balancing Finding

Since $\text{vol}_R^{\text{lower}}$ is a lower bound on vol_R , and $\text{vol}_R = \text{vol}_P$ restricted to the exercised sub-poset, the axiom inheritance analysis applies directly, with an additional layer: the lower bound itself is not a measure (it is a bound on a measure).

Theorem 3 (Axiom-Analogue Analysis for $\text{vol}_R^{\text{lower}}$). *$\text{vol}_R^{\text{lower}}$ is a lower-bound statistic, not a poset measure in the sense of [Lasser, 2026f, Proposition 1]; the clauses below describe operational analogues of M1–M6 for $\text{vol}_R^{\text{lower}}$, not literal axiom inheritance. M4 and the structural M6-failure result are substantive; M1, M2, M3, and M5 are weaker conditional statements noted for completeness.*

(M1) Non-negativity (analogue). *Satisfied.* $\text{vol}_R^{\text{lower}}$ is a sum of non-negative terms ($\Delta \text{vol}_P(X_n) \geq 0$ per event; max-attribution preserves non-negativity).

(M2) Null empty set (analogue). *Satisfied.* With zero sacrifice events, $\text{vol}_R^{\text{lower}} = 0$.

(M3) Monotonicity (conditional analogue). *Satisfied in the observational direction:* adding admissible sacrifice events weakly increases the constructed lower bound under Part A's subbatch-supremum form ($\text{vol}_R^{\text{lower}^A}$, Proposition 3) and under Part B's per-capability max-attribution form ($\text{vol}_R^{\text{lower}}$, Proposition 2). Both forms inherit monotonicity, with Part A applied to the subbatch supremum over admissible event subsets and Part B applied to the per-capability max over events covering each capability. M3 does not hold for the relationship between vol_P and $\text{vol}_R^{\text{lower}}$: adding a benchmarked capability to the poset increases vol_P but does not affect $\text{vol}_R^{\text{lower}}$ unless that capability is subsequently sacrificed. This is a lower-bound statistic property, not a poset-measure axiom.

(M4) Additivity under disjointness (substantive). *Satisfied under the parent-poset hypothesis.* By construction, vol_R is vol_P restricted to the exercised sub-poset $P^{\text{ex}} \subseteq P$. M4 for vol_P on P states that vol_P is additive over components of P that are pairwise poset-disjoint (no subsumption or cooperative-composition links). Take subsets $A, B \subseteq P^{\text{ex}}$ that are poset-disjoint in P —neither cooperative nor subsumption edges in the full poset connect any $a \in A$ to any $b \in B$, even via unexercised mediating capabilities. Note this is a stronger hypothesis than poset-disjointness in P^{ex} alone: the latter could permit a, b to share a parent-poset cooperative whose other participants happen to be unexercised in W , in which case A, B would be poset-connected in P even though neither cooperative nor subsumption edges restricted to P^{ex} connect them. Under poset-disjointness in P , M4 on P gives $\text{vol}_P(A \cup B) = \text{vol}_P(A) + \text{vol}_P(B)$; restricting both sides to P^{ex} gives $\text{vol}_R(A \cup B) = \text{vol}_R(A) + \text{vol}_R(B)$, establishing M4 for vol_R . Theorem 2's pairwise poset-independence hypothesis is stated against the parent poset P for this reason; the aggregate bound's M4 application uses parent-poset independence, not P^{ex} -internal independence.

(M5) Non-triviality (conditional analogue). A single sacrifice event with $\Delta \text{vol}_P(X) > 0$ produces $\text{vol}_R^{\text{lower}} > 0$ on the capabilities the event covers. This is an observation-dependent analogue, not the structural non-triviality of every valid capability.

(M6) Superadditivity under independence. The situation splits into a measure-axiom layer and a system-level layer:

- (a) vol_R as a measure on a fixed exercised sub-poset inherits M6. For any fixed P_W^{ex} , if $A, B \subseteq P$ are poset-independent in P , then $A \cap P_W^{\text{ex}}$ and $B \cap P_W^{\text{ex}}$ are poset-independent in P_W^{ex} , and vol_P 's M6 on P lifts to vol_R 's M6 on P_W^{ex} by the same restriction argument used for M4 above. This is literal axiom inheritance for the set function vol_R .

The structurally relevant failure is at the system level, not the measure-axiom level: vol_R is non-invariant under capability removal. Removing an unexercised capability c (a capability with $e_t(c) = 0$ throughout W) from the system leaves vol_R unchanged while vol_P drops by $w(c) > 0$. Example 3 witnesses this. The property is about the mapping $\mathcal{P} \mapsto \text{vol}_R^{\mathcal{P}}$ from capability systems to the measure, not about vol_R as a set function on any one P_W^{ex} ; it is what breaks the self-balancing transfer (Corollary 2).

- (b) $\text{vol}_R^{\text{lower}}$: **event-set non-additivity (a property of the lower-bound construction, not an M6 statement).** M6 is a measure axiom on poset-independent capability subsets, and applies to vol_R on a fixed exercised sub-poset by the restriction argument of (a). $\text{vol}_R^{\text{lower}}$ is not a measure of vol_R 's restriction; it is an observation-derived lower-bound functional of trade events. M6 is therefore not a property we have to inherit for $\text{vol}_R^{\text{lower}}$, and we make no claim that $\text{vol}_R^{\text{lower}}$ is super- or sub-additive on capability subsets. A separate property holds at the event-set level: merging two groups of sacrifice events changes per-capability attributions under max-attribution, and the joint maximum over all events can be less than the sum of within-group maxima. This is a non-superadditivity property of the lower-bound construction over event-set unions (Example 4 illustrates it in a constructed scenario).

outside Part B’s strict admissibility regime), retained for operational clarity about what max-attribution does on merged event groups. M4 still applies to $\text{vol}_R^{\text{lower}}$ on poset-independent capability subsets through the per-capability max-attribution sum: $\text{vol}_R^{\text{lower}}(A \cup B) = \text{vol}_R^{\text{lower}}(A) + \text{vol}_R^{\text{lower}}(B)$ when A, B are poset-disjoint, since the per-capability sums decouple over disjoint cooperative-closed subsets. M5 (every valid capability has positive weight) applies to vol_R on the exercised sub-poset by restriction, but $\text{vol}_R^{\text{lower}}$ assigns zero to any capability not covered by an admissible event, so $\text{vol}_R^{\text{lower}}$ is not an M5-style measure on the unexercised side.

Example 3 (vol_R is invariant under removal of unexercised capabilities). Consider a single-agent system with two poset-independent capabilities:

- $c_A = \text{“drive to gas station } \alpha\text{”}$, $w(c_A) = 1$.
- $c_B = \text{“drive to gas station } \beta\text{”}$ (different location), $w(c_B) = 1$.

The system poset is $P = \{c_A, c_B\}$ with $\text{vol}_P(P) = 2$. c_A and c_B are poset-independent: different fuel-access pathways with no shared participants.

Under the exercise indicator of Appendix A, the agent uses only c_A (gas station α is closer), so $P_W^{\text{ex}} = \{c_A\}$ and $\text{vol}_R(P) = \text{vol}_P(P_W^{\text{ex}}) = 1$. (Note: c_B is not load-bearing at any time in W —removing it leaves all realized outputs unchanged.)

Now remove c_B : the system becomes $P' = \{c_A\}$ with $\text{vol}_P(P') = 1$ (so vol_P contracts by $w(c_B) = 1$). The agent’s behavior is unchanged, so $P_W^{\text{ex}} = \{c_A\}$ and $\text{vol}_R(P') = 1$, unchanged.

The measure vol_R is therefore invariant under the removal of the unexercised c_B , while vol_P contracts by $w(c_B)$. A vol_R -maximizing actor would face no penalty for shedding unexercised capabilities; a vol_P -maximizing actor would pay $w(c_B)$.

The non-invariance lives at the level of the mapping $\mathcal{P} \mapsto \text{vol}_R^{\mathcal{P}}$: this mapping is not strictly monotone under capability-set expansion ($P \supset P'$ but $\text{vol}_R(P) = \text{vol}_R(P')$). Weak monotonicity ($P \supseteq P' \implies \text{vol}_R(P) \geq \text{vol}_R(P')$) holds; strict monotonicity fails, and the mapping is not strictly contracted by capability removal when the removed capability is unexercised. M6 itself remains satisfied on the fixed exercised sub-poset P_W^{ex} by restriction; the mapping-level non-invariance is what breaks the self-balancing transfer (Corollary 2).

Example 4 ($\text{vol}_R^{\text{lower}}$ non-additivity over event-set union). The example below illustrates that $\text{vol}_R^{\text{lower}}$ aggregated across two event groups is not in general the sum of the within-group aggregates. Max-attribution operates over events covering each capability rather than over poset-independent capability subsets, so the failure is on event-set merging specifically.

The example is constructed outside Part B’s strict admissibility regime: under Part B’s within-bundle poset-independence plus S5 plus the partition of unity $\sum_c \alpha_{n,c} = 1$, the share-collapse argument (Remark 1) forces $\alpha_{n,c}$ to equal the true share $\text{vol}_R(c)/\text{vol}_R(Y_n)$, leaving no room for two events on the same bundle to disagree on α . We relax those constraints below to expose the construction’s non-additivity property; the example illustrates the lower bound’s behaviour, not a Part-B-admissible scenario. Read it as “if a downstream variant of the framework permitted event-specific α on the same bundle (e.g., by dropping the partition of unity), aggregating $\text{vol}_R^{\text{lower}}$ across event groups would still fail super-additivity.”

Consider two sacrifice events with the same bundle $Y = \{a, b\}$ and (counterfactually) different decomposition coefficients:

- Event E_1 : $\Delta \text{vol}_P(X_1) = 1$, $\alpha_{1,a} = 0.5$, $\alpha_{1,b} = 0.5$.
- Event E_2 : $\Delta \text{vol}_P(X_2) = 1$, $\alpha_{2,a} = 0.3$, $\alpha_{2,b} = 0.7$.

Let group $A = \{E_1\}$ and group $B = \{E_2\}$. Under Theorem 2 Part B:

$$\begin{aligned} \text{vol}_R^{\text{lower}}(a \mid A) &= 0.5, & \text{vol}_R^{\text{lower}}(b \mid A) &= 0.5, \\ \text{vol}_R^{\text{lower}}(a \mid B) &= 0.3, & \text{vol}_R^{\text{lower}}(b \mid B) &= 0.7. \end{aligned}$$

Summing per-capability: $\text{vol}_R^{\text{lower}}(A) = 1.0$, $\text{vol}_R^{\text{lower}}(B) = 1.0$.

In the merged group $A \cup B$, max-attribution over both events: $\text{vol}_R^{\text{lower}}(a \mid A \cup B) = \max(0.5, 0.3) = 0.5$ and $\text{vol}_R^{\text{lower}}(b \mid A \cup B) = \max(0.5, 0.7) = 0.7$. Total: $\text{vol}_R^{\text{lower}}(A \cup B) = 1.2$.

If we (informally) compare a within-group sum $\text{vol}_R^{\text{lower}}(A) + \text{vol}_R^{\text{lower}}(B) = 2.0$ to the merged-event aggregate $\text{vol}_R^{\text{lower}}(A \cup B) = 1.2$, the merged aggregate is strictly smaller. This is not an M6 counterexample on capability subsets: A and B here are event sets, not poset-independent capability subsets, and M6 makes claims about the latter, not the former. What the example shows is that max-attribution does not commute with event-set union: merging two event groups whose attributions disagree on the same capability produces a per-capability max bounded above by the sum of within-group maxes. When restricted to the M6 setting (poset-independent capability subsets, with each capability covered by a single attribution event), this non-additivity does not arise; M6(b) below therefore fails for $\text{vol}_R^{\text{lower}}$ specifically through the event-set-union subadditivity exhibited here, not through any axiom-level failure on capability subsets.

This is a property of the lower-bound construction over event sets, not of the underlying vol_R : the true vol_R aggregate of the merged events need not be subadditive, but our observational lower bound is. The non-self-balancing corollary below therefore relies on the capability-removal non-invariance of M6(a), not on this lower-bound event-set subadditivity.

Corollary 2 (vol_R Is Not Self-Balancing). *The self-balancing property does not transfer from vol_P to vol_R . The GFM self-balancing theorem [Lasser, 2026f, Theorem 1] relies on the fact that any capability removal contracts vol_P , making vol_P -maximization incompatible with unilateral capability reduction. For vol_R , removing an unexercised capability leaves the measure unchanged (Example 3), so a vol_R -maximizer can shed unused capabilities without penalty. The protective coupling between measure-maximization and capability preservation that vol_P 's self-balancing encodes therefore does not apply to vol_R .*

The corollary does not follow from vol_R failing M6 as a measure axiom—it does not (Theorem 3, M6(a))—nor from $\text{vol}_R^{\text{lower}}$ failing M6 alone (a lower bound failing measure axioms is unremarkable). The corollary rests on the system-level non-invariance of M6(a): vol_R 's value depends on which capabilities are exercised, and unexercised capabilities can be removed without affecting that value.

This is the structural reason to use vol_R as a diagnostic rather than an objective. An actor maximizing vol_R would lack the automatic diversity-preservation that makes vol_P safe: exercise is rewarded, but dormant optionality is not protected.

Remark 16 (When the operational analogues of M1–M6 hold for $\text{vol}_R^{\text{lower}}$). *The event-set non-additivity of (b) above does not arise when the merged groups have sacrifice events in poset-independent bundle categories (Definition 10): each unbenchmarked capability appears in sacrifice events from only one group, and cross-group capabilities are poset-independent. Under this non-redundancy condition, $\text{vol}_R^{\text{lower}}$ satisfies the operational analogues of M1, M2, M3, M4, and (the M6-analogue stated in (b)) consistently; $\text{vol}_R^{\text{lower}}$ remains a lower-bound functional, not a poset measure, so this is operational compatibility rather than literal axiom inheritance. Even so, the self-balancing property transfers to $\text{vol}_R^{\text{lower}}$ only as a lower bound on self-balancing— $\text{vol}_R^{\text{lower}}$ being self-balancing does not imply vol_R itself is self-balancing.*

6 Commitment-Layer Composition

6.1 Motivation

The revealed-sacrifice channel observes trade events. In its raw form, each event exposes (i, X, Y, t) —the agent's identity, what they surrendered, what they acquired, and when. This is already more private than direct vol_R measurement, but the framework can do better: the commitment protocol of [Lasser, 2026a, Definition 6] allows the sacrifice event to be committed with hiding, so that only the vol_R -relevant content (the magnitude $\Delta \text{vol}_P(X)$ and the category of Y) is revealed.

6.2 Formal construction

Definition 9 (Committed Revealed-Sacrifice Event). *A committed revealed-sacrifice event is a tuple (Γ, π, t) where:*

- $\Gamma = \text{Commit}((\Delta\text{vol}_P(X), \text{category}(Y)), r)$ is a commitment to the sacrifice magnitude and bundle category, using randomness r [Lasser, 2026a, Definition 6];
- π is a zero-knowledge proof [Lasser, 2026a, Definition 7] that:
 - (a) the commitment Γ binds the objectively verifiable components of the underlying record: the objective component of $S2$ ($\Delta\text{vol}_P(X)$ correctly computed from X via the public benchmark) and the bundle-containment check (item (d) below). “Valid revealed-sacrifice event” in the strict sense (all of $S0$, $S1$, $S2$, $S3$, $S4(a)$, and the genuine-exchange conditions) is not certified by this proof; the proof attests only the objective magnitude and containment fields, leaving the substantive admissibility hypotheses to external attestation (Remark 18) or richer commitments (Remark 20, Remark 17);
 - (b) $\Delta\text{vol}_P(X) \geq 0$;
 - (c) the trade actually occurred (linked to a verifiable ledger entry);
 - (d) the bundle Y lies wholly within the claimed partition component: $Y \subseteq C_{\text{category}(Y)}$ (Definition 10(ii));
- t is the event timestamp (public).

The committed event reveals $\Delta\text{vol}_P(X)$ (the sacrifice magnitude) and $\text{category}(Y)$ (the bundle’s partition label). It hides agent identity i , the specific benchmarked capability X , the specific unbenchmark bundle Y , and the counterparty. Price hiding caveat (money channel). Because the benchmark function on purchasing power (Definition 7) is publicly known and typically monotone in price, the committed $\Delta\text{vol}_P(X)$ is informative about p : a verifier with the benchmark function can invert Δvol_P to recover p for the surrendered purchasing power. The commitment hides the counterparty of the trade and the specific unbenchmark bundle Y , not the price; the “price hiding” phrasing in informal summaries should be read as “hides what was bought from whom,” not “hides what was paid.” Where stronger price hiding is required (e.g., trade volumes that would identify the agent), the hiding variant of Proposition 5 additionally hides $\Delta\text{vol}_P(X)$ itself, releasing only an aggregate threshold B .

Remark 17 ($S0$, $S1$, $S3$, $S4(a)$, and $S5$ are not ZK-verifiable under the default commitment). Assumptions $S0$ (calibration), $S1$ (free choice), $S3$ (bundle coherence), $S4(a)$ (valuation additivity), and $S5$ (decomposition validity) are about the agent’s internal state, the true vol_R distribution, or the bundle’s hedonic decomposition. These cannot be verified in zero knowledge [Goldwasser et al., 1985] via the default committed event of Definition 9 without an oracle for the agent’s subjective state or richer committed fields exposing the decomposition. The default ZK proof verifies the objective magnitude conditions ($S2$ ’s benchmark computation) and cross-category poset-independence (which is the part of $S4(b)$ needed for Part A aggregation across distinct partition components, via Definition 10(i)); within-bundle and per-capability $S4(b)$ facts needed for Part B’s per-capability attribution are not certified by the default commitment. The remaining conditions are structural assumptions about the trade environment or per-bundle decomposition, not properties the default commitment protocol can enforce. $S3$ in particular requires either external attestation that the bundle’s decomposition is non-degenerate, or a richer commitment that exposes per-bundle decomposition fields and proves their full-rank property in ZK; cf. Remark 20.

Definition 10 (Category Partition). *The function $\text{category}(Y)$ maps each unbenchmark bundle to a label in a public partition $\mathcal{C} = \{C_1, \dots, C_m\}$ of the unbenchmark capability space U , subject to two conditions:*

- (i) Poset-independence: for any $j \neq k$, no capability in C_j is related by subsumption or cooperative composition to any capability in C_k .
- (ii) Bundle containment: each bundle Y lies wholly within a single partition component: $\text{category}(Y) = j$ iff $Y \subseteq C_j$.

The partition is published as part of the framework’s measurement infrastructure and is fixed for a given trade window. This definition is load-bearing for aggregation: distinct category labels imply poset-independence of the underlying bundles (Remark 11), which is what Theorem 2 Part A requires.

6.3 Computability of the Part A formula from committed events

Proposition 4 (Part A Formula is Computable from Committed Events on Distinct-Category Batches). *For a batch of committed revealed-sacrifice events (Definition 9) whose revealed category labels $\{\text{category}(Y_n)\}$ are pairwise distinct, the formula of Theorem 2 Part A (the right-hand sum $\sum_n \Delta \text{vol}_P(X_n)$ together with the poset-independence check between bundles) is evaluable without opening the hidden fields: the magnitudes $\Delta \text{vol}_P(X_n)$ and labels $\text{category}(Y_n)$ are revealed, and distinct labels imply poset-independent bundles via Definition 10(i). Batches with colliding labels fall outside this certificate and must be handled separately (Remark 19).*

This is a computability result, not an admissibility result. Applying Theorem 2 Part A to a batch of committed events additionally requires that the theorem’s Part A hypotheses hold: the genuine-exchange conditions (i)–(iv) of Definition 2 and assumptions S0, S1, S2, and S4(a). The ZK proof π of Definition 9 certifies only the objectively verifiable subset (S2’s magnitude component and the bundle-containment check $Y_n \subseteq C_{\text{category}(Y_n)}$). The remaining Part A hypotheses (S0 valuation-bounded-by-exercise, S1 free choice, S4(a) valuation additivity, and the cooperative-closure conditions (iii)–(iv)) are about agent-internal state or poset-structural facts the default commitment protocol cannot itself attest (Remark 17). Part B additionally requires S3 (bundle coherence), which is not certified by the default commitment either; Part B’s per-capability use of S3 inherits this dependency. Their validity must be established externally: by institutional filters on the trade environment, by sampled audit, or by assumption at the batch level.

Proof. We establish formula computability; admissibility is discussed in the statement and is not claimed here.

(i) *Sacrifice magnitudes.* By Definition 9, each committed event reveals $\Delta \text{vol}_P(X_n)$ directly, so the aggregate $\sum_n \Delta \text{vol}_P(X_n)$ is evaluable without opening the commitment. (In the hiding variant of Proposition 5, only B and a ZK attestation $\sum_n \Delta \text{vol}_P(X_n) \geq B$ are published; both forms expose the aggregate at the intended privacy level.)

(ii) *Pairwise poset-independence of bundles (distinct labels).* By Definition 10(i), any two distinct partition components C_j and C_k are poset-independent: no subsumption or cooperative-composition link connects any $c \in C_j$ to any $c' \in C_k$. Definition 9(d)’s ZK proof attests that $Y_n \subseteq C_{\text{category}(Y_n)}$, so Y_n lies wholly within a single component. Under the pairwise-distinct-label hypothesis, the bundles lie in distinct components and inherit poset-independence from the partition. Both checks use only revealed or ZK-attested fields.

When two events share the same category label $\text{category}(Y_n) = \text{category}(Y_{n'})$, both bundles lie inside the same partition component C_j ; poset-independence between $Y_n, Y_{n'} \subseteq C_j$ is a finer fact about sub-structure within C_j that Definition 9 does not attest. The distinct-label certificate is therefore a *sufficient* certificate for Part A’s poset-independence hypothesis, not a *necessary* one; see Remark 19 for how to handle batches with colliding labels.

(iii) *What is not certified.* Neither the above computation nor the ZK proof π establishes the genuine-exchange conditions (iii)–(iv) (cooperative-closure across $X/H \setminus X$ and $Y/H \setminus X$), S0 (agent’s valuation bounded by realized exercise), S1 (voluntariness and non-degradation), or S4(a) (valuation additivity over bundles). These depend on facts about the agent’s internal state, the pre-trade holding set, and the poset structure that are not exposed by the commitment. The Part A bound is admissible only when those are established by an external mechanism (Remark 18). ■

Remark 18 (External attestation model). *The aggregate lower bound theorem separates two kinds of facts: those a ZK proof on committed events can certify, and those it cannot. This remark characterizes the second class, the infrastructure required to attest to it, and the framework’s graceful-degradation behaviour when that infrastructure is partial.*

1. What makes a property non-ZK-verifiable. *A property is ZK-verifiable from a committed event when it is a function of the cleartext fields of that event (magnitude, category, timestamp)*

together with public structural inputs (the poset P , the category partition C , the benchmark function on vol_P). It is non-ZK-verifiable when its truth depends on data outside this surface:

- (i) Agent-internal state not encoded in the event tuple: the agent’s subjective valuation function U_i and its scale alignment with vol_R ($S0$); the additivity structure of U_i over bundles ($S4(a)$); the agent’s choice set A_i and which alternatives in it are non-degrading ($S1$).
- (ii) Counterfactual claims about unrealized behaviour: whether the trade would have happened under coercion or duress, whether retention would have been degrading, whether a third non-degrading alternative was actually available.
- (iii) Per-agent global state not exposed by any single event: the agent’s pre-trade holdings $H_i \setminus X$ used in the genuine-exchange conditions (iii)–(iv) of Definition 2, which quantify over the cooperative/subsumption structure spanning X and $H_i \setminus X$.

$S0$, $S1$, and $S4(a)$ fail criterion (i); coercion/non-degradation attestations fail (ii); the genuine-exchange retained-holdings conditions fail (iii). The companion paper [Lasser, 2026d, Need Sufficiency]’s sufficiency thresholds, downstream-safety adjudication, and residual-class classification fail (i) and (ii) similarly. Every assumption the commitment layer cannot certify is non-ZK-verifiable for at least one of these reasons.

2. What constitutes an “institutional layer.” An institutional layer is any infrastructure that produces attestations: signed claims, with audit trail, that bind a named attesting authority to a specific assertion about a non-ZK-verifiable property. An attestation protocol comprises (i) credentialing of attesting authorities (who is permitted to attest); (ii) attestation content (what is asserted, how it is bound to specific events or batches); (iii) audit infrastructure (sampled records, dispute resolution, evidence retention); (iv) consequences for false attestation that make the protocol incentive-compatible in deployment. The framework does not legislate any specific institutional design; it requires that attestations of the types catalogued in 3 below are produced with reliability adequate to the diagnostic’s stated confidence level. Plausible institutional realizations span regulatory bodies, auditors, multi-stakeholder governance councils, and substrate-internal compliance functions; selection is a deployment-context decision.

3. Required versus degraded operation. The attestation outputs the framework consumes are catalogued in four layers, each with a different attestation locus and each with a defined behaviour when partially supplied:

- (A) $S1$ -admissibility labelling. Every event entering the aggregate must carry an attested label asserting $S1$ holds (some non-degrading alternative to the trade existed in A_i). The $H1+H2$ below-sufficiency sub-class of the companion paper’s $S1$ -sufficiency definition [Lasser, 2026d, Need Sufficiency] is the structurally-diagnosable sub-case; below-sufficiency targeting with an attested third non-degrading alternative may still be admitted; events failing both routes are filtered.
- (B) Public-poset curation. The category partition C must be curated for cross-component poset-independence (Definition 10(i)). This is a governance act on a public structural artefact, not a per-event attestation; once curated, the property is ZK-verifiable for any committed event whose label opens to a valid leaf of C .
- (C) Per-event genuine-exchange attestations. The retained-holdings conditions (iii)–(iv) of Definition 2 require trade-level attestation by the agent, the counterparty, or sampled audit of trade records. This is the per-event analogue of (B): unlike public-partition curation, retained-holdings facts are agent-specific and cannot be certified by curating C .
- (D) Batch-level $S0$ and $S4(a)$ reconciliation. $S0$ (calibration) and $S4(a)$ (valuation additivity) are assumed at the batch level with periodic sampled-audit reconciliation against external evidence (longitudinal exercise studies, market-segment behavioural analysis, controlled-relaxation testing of the calibration chain).

Required for the theorem to apply at full strength is that all four layers produce their attested outputs at the aggregate’s stated confidence level. Degradation is the framework’s graceful-degradation behaviour when an attestation layer is partial:

- *Missing layer-(A) labels filter events conservatively (excluded from the aggregate); the lower bound holds on the attested subset and shrinks as the unlabeled fraction grows.*
- *Missing layer-(C) attestations on a fraction of events filter those events; the bound holds on the attested remainder.*
- *Missing layer-(D) batch reconciliation does not immediately invalidate the bound but means the deployment cannot certify $S0/S4(a)$ hold; the diagnostic should be reported with the unverified- $S0/S4(a)$ caveat, and the bound should be read as conditional on the un-attested assumptions.*
- *Missing layer-(B) curation invalidates the partition's cross-component independence and disables Part A's category-aggregation route on the affected partition; Part B per-capability aggregation may remain available where its hypotheses hold per event.*

The graceful-degradation property is one-sided: missing attestation always shrinks (or invalidates) the lower bound, never inflates it. The framework cannot produce false alarms from missing attestation; it can produce under-counted bounds that fail to fire alarms that the truth would justify.

4. What this pair of papers calls for. *This pair specifies what the institutional layer must produce (Layers A–D above) and how the framework behaves when those outputs are partial. It does not specify how the attestation protocol should be structured to produce those outputs reliably; protocol design is substrate- and jurisdiction-specific, and a national-scale governmental deployment looks different from a single-organization deployment. The natural follow-up is a dedicated treatment of the attestation protocol covering credentialing, attestation content, audit cadence, dispute resolution, and adversarial robustness. Pending that follow-up, deployments of the sacrifice channel and its diagnostic architecture should be read as contingent on the attestation layer's practical adequacy in the specific deployment context, with the graceful-degradation property above as the formal floor.*

Remark 19 (Handling category-label collisions). *When two committed events share a category label, Proposition 4's distinct-label certificate does not apply to that pair: both bundles lie in the same partition component C_j , and poset-independence between same-component subsets is finer structure than the commitment interface exposes. Three operational responses are available without weakening the theorem:*

- Sub-batch partitioning (caveat: not generally summable). Partition the batch into maximal sub-batches whose labels are pairwise distinct, and compute Part A separately per sub-batch; each sub-batch's lower bound is valid for its own covered subset. Summing across sub-batches to obtain a full-batch aggregate is not generally licensed: when the same category label appears in multiple sub-batches (the typical case for a colliding batch), the sub-batches share that category's bundles, so adding their Part-A sums double-counts the shared category. The full-batch aggregate via sub-batch sum is licensed only if the sub-batches partition by distinct categories (i.e., each category label appears in exactly one sub-batch), which holds trivially when the original batch had no collisions. For genuinely colliding batches, fall back to Part-A's sub-batch supremum (Proposition 3): the tightest Part A bound is the supremum over admissible sub-batches, not the sum across them.*
- Finer partition. Refine the public partition C to sub-components such that typical batch collisions become distinct-label pairs. This is a governance choice trading category-granularity against commitment complexity.*
- Richer commitments. Extend Definition 9 to attest explicit subset-disjointness or cooperative-closure of $Y_n, Y_{n'}$ within their shared component C_j . This is the path Part B's per-capability certificate (Remark 20) also walks.*

The paper's stated results use option (a) implicitly: Proposition 5's ZK-aggregation rollup explicitly proves pairwise-distinct labels as part of its (ii), matching the distinct-batch certificate of Proposition 4.

Remark 20 (Part B requires additional committed fields). *Part B's per-capability refinement requires $\alpha_{n,c}$ coefficients and stable capability identities across events. The committed interface reveals only category labels. For Part B to operate on committed data, the commitment must addi-*

tionally expose a per-capability coordinate system within each category component and the corresponding α values. This is a richer commitment that reveals more structure than Part A requires.

Proposition 5 (ZK Aggregation over Hiding-Committed Events). *The Pedersen commitment scheme underlying Definition 9 is hiding [Lasser, 2026a, Definition 6] by default; the accompanying disclosure of $(\Delta\text{vol}_P(X_n), \text{category}(Y_n))$ is an optional publication choice. Consider the hiding variant of the committed-event interface in which each event n publishes a pair of separate Pedersen commitments $(\Gamma_n^\Delta, \Gamma_n^{\text{cat}})$ to $\Delta\text{vol}_P(X_n)$ and to $\text{category}(Y_n)$ respectively, together with a rollup-stage tuple-binding proof linking the two commitments to the same event n , and the rollup-stage proof π described below. Per-event timestamps t_n are not published individually; the rollup publishes only the aggregate window $W = [t_0, t_0 + T]$ within which all events fall. (Publishing per-event timestamps would leak the true batch size N via a count of timestamp tuples; the hiding variant therefore aggregates timestamps to the window level.)*

Fix a public upper bound $N_{\max}$ on batch size. Given a batch of at most $N_{\max}$ hiding-variant committed events, a rollup proof π can establish:

- (i) $\sum_n \Delta\text{vol}_P(X_n) \geq B$ (aggregate bound on magnitudes alone, using the homomorphism on the magnitude commitments $\{\Gamma_n^\Delta\}$);
- (ii) the non-null bundle category labels $\{\text{category}(Y_n) : \text{category}(Y_n) \neq \text{null}\}$ committed by $\{\Gamma_n^{\text{cat}}\}$ are pairwise distinct (i.e., among the $N \leq N_{\max}$ real events, no two share a category label; the $N_{\max} - N$ null-padded positions are excluded from the pairwise-distinct check, so multiple null entries do not violate the condition);
- (iii) each Γ_n^{cat} commits to a label in the public partition $\mathcal{C}$ (Merkle membership), and is bound to the same event n as Γ_n^Δ via the tuple-binding proof;

for a public bound B , without revealing the individual $\Delta\text{vol}_P(X_n)$, the specific category labels, or the true batch size $N \leq N_{\max}$. Separating the two commitments is what isolates the magnitude sum from the label-vector constraint: the homomorphic product $\prod_n \Gamma_n^\Delta$ binds only to $\sum_n \Delta\text{vol}_P(X_n)$, and the label-vector proof on $\{\Gamma_n^{\text{cat}}\}$ operates on a disjoint commitment set.

By Definition 10, distinct category labels correspond to poset-independent partition components, so (ii) establishes the poset-independence required by Theorem 2 Part A. Both (i) and (ii) are formula inputs that the ZK rollup certifies; the theorem’s admissibility hypotheses (the genuine-exchange conditions of Definition 2 and assumptions S0, S1, S2, S4(a)) remain outside the ZK layer (Remark 17) and must be supplied through external attestation (Remark 18).

Proof. The prover holds the full set of N committed events together with their openings $(\Delta\text{vol}_P(X_n), \text{category}(Y_n), r_n^\Delta, r_n^{\text{cat}})$. To standardize batch size the prover pads with $N_{\max} - N$ “null” events for which the magnitude commitment Γ_n^Δ commits to 0 and the category commitment Γ_n^{cat} commits to the distinguished null label, both with fresh randomness. Padding is homomorphically invisible on the magnitude side (null events contribute 0 to the magnitude sum), and the null label is excluded from the non-null pairwise-distinct check on the category side.

Component (i)—range proof on the aggregate magnitude sum. Pedersen commitments operate on a finite-field representation of the magnitudes; we assume a domain-supplied fixed-point encoding with scale σ and per-event upper bound $\Delta_{\max}$ such that each $\Delta\text{vol}_P(X_n)$ is encoded as a non-negative integer in $[0, \lfloor \sigma \cdot \Delta_{\max} \rfloor]$ via floor rounding (encoded magnitude $\leq \sigma \cdot \text{true magnitude}$), and the aggregate fits in $[0, N_{\max} \cdot \lfloor \sigma \cdot \Delta_{\max} \rfloor] \subset \mathbb{Z}_p$ for the curve order p . Floor rounding gives, per event, $\text{encoded}_n \leq \sigma \cdot \text{true}_n$, hence summing, $\text{encoded_sum} \leq \sigma \cdot \text{true_sum}$, equivalently $\text{true_sum} \geq \text{encoded_sum} / \sigma$. This direction of the floor-rounding inequality is *conservative for lower bounds on true_sum*: a range-proof on the encoded sum certifying “ $\text{encoded_sum} \geq \sigma B$ ” therefore implies $\text{true_sum} \geq B$ directly, with *no* rounding margin needed in this direction (the $N_{\max} / \sigma$ slack runs the other way and would matter only for upper-bounding true_sum , which the rollup does not do). Using a standard Pedersen commitment range proof [Lasser, 2026a, Definition 7], the prover demonstrates $\sum_{n=1}^{N_{\max}} \Delta\text{vol}_P(X_n) \geq B$ in the true-sum units by certifying $\text{encoded_sum} \geq \sigma B$ in encoded units. The magnitude-summation commitment is the product $\prod_n \Gamma_n^\Delta$, which binds to $\sum_n \Delta\text{vol}_P(X_n)$ by the additive-homomorphic property of Commit on the magnitude commitments. Because Γ_n^Δ is committed independently of Γ_n^{cat} , the homomorphic product does not mix magnitudes with category labels. Null events contribute 0 so the sum equals

$\sum_{n=1}^N \Delta \text{vol}_P(X_n)$. The proof reveals neither the individual $\Delta \text{vol}_P(X_n)$ nor the true N (the verifier sees only a length- $N_{\max}$ vector of magnitude commitments).

Component (ii)—pairwise-distinct non-null labels. The prover operates on the existing category commitments $\{\Gamma_n^{\text{cat}}\}_{n=1}^{N_{\max}}$ (no new commitment to a separate label vector is needed; the rollup binds to the already-committed Γ_n^{cat} values). The ZK argument shows (a) each non-null category (Y_n) opens to a valid leaf of the public partition Merkle root $\mathcal{C}$, and (b) the multiset of non-null labels contains no duplicates. Duplicate-freeness is a standard ZK set-disjointness argument: commit to a sorted copy, prove strict-monotone ordering among non-null elements, and prove the permutation relation to the unsorted commitments via a shuffle argument [Lasser, 2026a, Definition 7].

Component (iii)—tuple binding. For each n , a tuple-binding proof links Γ_n^Δ and Γ_n^{cat} to the same underlying event, preventing label-magnitude reshuffling across events. Standard Pedersen-commitment-based tuple-binding suffices: the prover commits to a sequencing tag and proves both commitments share the tag.

Formula-input certification. By Definition 10(i), pairwise-distinct non-null labels imply pairwise poset-independent bundles. Combined with (i), the rollup certifies the two ZK-verifiable inputs to the Part A formula (Proposition 4): the aggregate sacrifice magnitude meets the threshold B , and the bundles are pairwise poset-independent. The verifier learns only the public parameters $(B, N_{\max}, \mathcal{C})$ and the length- $N_{\max}$ commitment vector.

Admissibility is not ZK-certified. The inference $\text{vol}_R^{U, [W]} \geq B$ requires, in addition, that Theorem 2 Part A’s non-ZK-verifiable hypotheses hold across the batch: the genuine-exchange conditions (i)–(iv) of Definition 2, S0, S1, S2, and S4(a). These are not part of the rollup proof (Remark 17). Under external attestation of those hypotheses (Remark 18), the hidden-batch rollup certifies the existential statement “there exists a subset $U \subseteq P$ of unbenchmarked capabilities, drawn from the public partition $\mathcal{C}$ and covered by this batch’s hidden bundles, on which $\text{vol}_R^{U, [W]} \geq B$.” U itself is not exposed by the hiding-variant rollup—the verifier learns the threshold B and the existence of such a U , but not which categories of $\mathcal{C}$ comprise it. Without external attestation, the rollup certifies only that the Part A formula’s inputs meet the stated thresholds while hiding individual $\Delta \text{vol}_P(X_n)$, specific category labels, the true N , and the realized subset U . ■

Remark 21 (Hiding variant is strictly richer than Definition 9). *The hiding-variant interface is a strict privacy upgrade: it hides what Definition 9 discloses. The default interface suffices for routine operational accounting where committed events are published to a public ledger; the hiding variant is useful when a higher-level aggregator publishes only batch-level attestations to downstream verifiers. Both interfaces rely on the same Pedersen commitment primitive; they differ only in which openings are released alongside the commitment.*

7 Integration: Duality and the Shared Privacy Discipline

7.1 The complementary pair

	Controlled relaxation [Lasser, 2026h, Controlled Relaxation]	Revealed sacrifice (this paper)
Question	Is the restriction warranted?	Is the capability valuable?
Evidence	Framework-initiated exercise	Agent-initiated sacrifice
Direction	Framework injects observation	Framework receives disclosure
Scope	Bounded sub-poset S , bounded τ	Bounded to the trade event
Privacy cost	Scope-limited monitoring	Magnitude/category disclosure (default); price-invertibility through the public benchmark
Adversarial model	Agent must tolerate test	Agent must operate (buy, work)
Dimension	Risk (restriction lattice)	Value (experiential vol_R)

Together they cover both axes of the preemptive-restriction criterion of [Lasser, 2026c, Proposition 1]: restriction cost (controlled relaxation tests the risk side) and experiential value (revealed sacrifice bounds the value side). Controlled relaxation tells us whether a feared harm is real; revealed

sacrifice tells us whether the forgone capability is worth recovering. Without both, the restriction decision is underdetermined.

7.2 The shared privacy discipline

Three papers in the sequence [Lasser, 2026e,h] and the present paper share a structural discipline: **resolution through frequency, not through depth.**

- The phase boundary of [Lasser, 2026e, Theorem 2]: $\rho_{\min}^{\text{cross}}$ counts observation *channels*, not depth of observation per channel.
- Controlled relaxation [Lasser, 2026h, Controlled Relaxation]: more precision on risk claims comes from running more tests (Lasser, 2026h, Theorem 2, convergence depends on test count), not from surveilling each test more intrusively.
- Revealed sacrifice (this paper): more precision on vol_R comes from observing more trades (Propositions 3 and 2), not from observing each trader more intrusively. The commitment layer (Section 6) reduces the trade-level observation to the formal privacy properties of Theorem 1; the operational “privacy-minimal” characterization additionally requires the institutional preconditions of Proposition 1.

This shared discipline is the sequence’s answer to the panopticon temptation: the framework gains measurement capability through *more independent signals*, not through *deeper access to any single signal*. This is consonant with the cross-substrate redundancy principle of [Lasser, 2026e, Theorem 2] and the cryptographic opacity commitment of [Lasser, 2026a, Definition 6].

7.3 Connection to the phase-boundary analysis

The phase-boundary analysis of [Lasser, 2026e, Phase Redundancy] assumes vol_P accurately measures the quantity the framework optimizes. The present paper asks whether that assumption holds. When β^{lower} is low, the phase-boundary parameters (r_S, r_W from [Lasser, 2026e, Theorem 1]) may be miscalibrated because the actor is optimizing a proxy (vol_P) that diverges from the true target.

A natural extension (not pursued here): extend the phase-boundary analysis to incorporate the $\text{vol}_P/\text{vol}_R$ divergence. The effective self-correction rate r_S^{eff} would depend on β^{lower} .

7.4 Architecture layers developed in the companion paper

This paper establishes the lower-bound measurement theorem as a self-contained result. The companion paper [Lasser, 2026d, Need Sufficiency] is a downstream work that builds three architectural layers atop the theorem proved here:

- **Operationalization of S1’s non-degradation primitive.** This paper states S1 with “below-sufficiency” as a primitive (Section 2.1) and assumes an external filter rejects S1-failing events from the aggregate (in particular, the H1+H2 below-sufficiency sub-class where retention is degrading and no third non-degrading alternative is available). The companion paper [Lasser, 2026d, Need Sufficiency, Section 2] gives one structural implementation of that diagnosis via multi-dimensional need bundles, downstream-safe sufficiency, polarity-correct benchmarks, and downstream-gated weighting.
- **Gap decomposition and diagnostic interpretation.** Theorem 2 lower-bounds $\text{vol}_R^{U,[W]}$ on the observed subset; interpreting *why* β^{lower} is low for a given window requires partitioning the gap into restricted, covered, dormant, residual, and (under the default literal-partition strategy) boundary-residual contributions. The companion paper [Lasser, 2026d, Need Sufficiency, Section 3] supplies the five-cell decomposition (default strategy (c), identity by atomic-weight accounting over disjoint cells), a four-cell variant under cooperative-closure strategies (a)/(b), and an $O(|P| + E + M)$ computability proposition.
- **Wireheading-consistent signature and residual class.** The companion paper [Lasser, 2026d, Need Sufficiency, Sections 4–5] develops a Herfindahl–Hirschman-Index-based concentration signal from trade-flow data—Schur-convex in the majorization order, with a Part-A category-flow variant third-party observable from public committed events plus the

public S1-admissibility labelling—and characterizes the residual class (capabilities structurally outside the sacrifice channel’s reach) via a revealed-observation curve.

The present paper’s results stand independently of the companion paper: the proofs do not invoke any definition developed in [Lasser, 2026d, Need Sufficiency]. Readers interested only in the measurement theorem and its commitment composition can stop here; readers interested in the structural filter, gap decomposition, and wireheading-consistent diagnostic should proceed to [Lasser, 2026d, Need Sufficiency].

8 Discussion and Open Questions

8.1 What this paper establishes

Global vol_R measurement is privacy-incompatible under the framework’s existing commitments. The framework’s commitments in [Lasser, 2026a,e,h] rule out the panopticon-style observation that direct global measurement would require. Within a consented operational perimeter, exact direct measurement remains available (Appendix A); the privacy-incompatibility claim is about out-of-perimeter and global measurement.

Revealed sacrifice is a privacy-respecting surrogate on the observed subset. The sacrifice channel satisfies three formal privacy properties (Theorem 1) plus two institutional preconditions (Proposition 1), and composes with the commitment layer of [Lasser, 2026a, Exogenous Verification] (Propositions 4 and 5) to produce an observation mechanism on distinct-category batches whose privacy profile is characterized by the formal-plus-institutional split above: the formal properties are theorem consequences, and the “privacy-minimal” summary phrasing is conditional on the institutional preconditions of Proposition 1 holding operationally.

The B-to-C gap is partially constrained by trade data. The aggregate lower bound (Theorem 2) constructs the sacrifice-channel ratio β^{lower} , which (under the denominator-and-domain conventions of Definition 6) is a lower bound on the true exercised fraction β when the same denominator is used— $\beta^{\text{lower}}(U, T) \leq \beta_U$ on the observed subset U , and $\beta^{\text{lower}}(P, T) \leq \beta_P$ on the global denominator P under the zero-extension convention for capabilities outside U . Equivalently, $1 - \beta^{\text{lower}}$ upper-bounds the correspondingly-defined B-to-C gap. Both inequalities hold under S0–S2, S4(a), and the genuine-exchange conditions of Definition 2. β^{lower} is the object the framework computes; β is the latent quantity bounded indirectly through it. The lower bound is silent on capabilities outside the observed subset and on the residual class of [Lasser, 2026d, Need Sufficiency]; it does not characterize the B-to-C gap globally, only constrains the gap from above ($1 - \beta \leq 1 - \beta^{\text{lower}}$) on what trade data covers. The constructed lower bound is non-decreasing in observed admissible events: Part A via sub-batch supremum (Proposition 3) and Part B via per-capability max-attribution (Proposition 2).

Two sacrifice channels, substitutable with residuals. The money and time channels are substitutable via cooperative markets (employment and hiring) and triangulate the same underlying sacrifice substrate. Under the cross-channel closure and admissibility hypotheses of Corollary 1, the joint aggregate weakly dominates either channel alone; strict domination is *available* at the non-fungibility residuals (agent-specific time, capital-concentrated access) when the trade window contains at least one admissible positive-magnitude residual-category event poset-independent from the single-channel maximizing sub-batch (the corollary’s qualifications, which the discussion summary does not restate—see Corollary 1 for the full conditions). “Weakly dominates” here is a corollary statement, not an unconditional claim: without the closure hypothesis (cross-channel pairwise poset-independence), Part A’s M4 step does not apply across the two channels’ bundles and the weak-domination claim does not follow.

vol_R is not self-balancing. vol_R exhibits a system-level non-invariance under capability removal: eliminating an unexercised capability leaves vol_R unchanged while vol_P contracts (Example 3). This is a property of the mapping $\mathcal{P} \mapsto \text{vol}_R^{\mathcal{P}}$ from capability systems to the measure, separate from M6 on any fixed exercised sub-poset (Theorem 3). The self-balancing protection vol_P enjoys

(every removal contracts the measure) therefore does not transfer to vol_R (Corollary 2). vol_R is a diagnostic, not an objective. This structural finding is independent of the observation mechanism.

Complementarity with controlled relaxation. Together, this paper and [Lasser, 2026h, Controlled Relaxation] cover the risk and value dimensions of the preemptive-restriction criterion of [Lasser, 2026c, Proposition 1]. Neither is sufficient alone.

8.2 Limitations

Assumption load. The full per-capability refinement (Theorem 2 Part B) requires S0–S5 plus within-bundle poset-independence (Definition 4). S0 (calibration) and S1 (free choice) are the most empirically demanding, and neither is ZK-verifiable (Remark 17). Part A (S0–S2 and S4(a) only) is the more defensible result; Part B should be understood as the refinement available under stronger conditions.

B-to-C gap inheritance. The damage bound of [Lasser, 2026h, Theorem 1] bounds contraction of vol_P , not of the experiential target directly. The present paper provides a lower bound on vol_R as a diagnostic, but the diagnostic is itself subject to the B-to-C gap on the benchmarked side: the sacrifice signal $\Delta \text{vol}_P(X)$ is only as accurate as vol_P ’s measurement of the benchmarked capability.

External admissibility layer. Several non-ZK-verifiable hypotheses require institutional attestation to hold operationally: S0 (valuation bounded by exercise), S1 (free choice and non-degradation), S4(a) (valuation additivity), and the genuine-exchange conditions (iii)–(iv) of Definition 2. Remark 18 describes the operational shape of this attestation layer. The present paper’s theorems hold under any realization of the admissibility layer that correctly identifies S1-failing events—in particular, the below-sufficiency sub-class where retention is degrading and no third non-degrading alternative exists—and rejects them from the aggregate; one structural implementation of the diagnosis for that sub-class is given in the companion paper [Lasser, 2026d, Need Sufficiency].

8.3 Open questions

1. **Hedonic disaggregation of bundles.** Theorem 2 Part B requires event-local decomposition coefficients $\alpha_{n,c}$. Classical hedonic regression [Rosen, 1974] handles the money channel. The time-sacrifice channel needs an analogous framework: how does one decompose a block of time spent on a complex activity (parenting, which involves teaching, emotional regulation, physical care, relationship-building) into per-capability contributions?
2. **Mixed-polarity bundle-for-bundle trades.** Bundle-for-bundle trades are formally supported by Definition 2 (cf. Remark 5), but events whose surrendered and acquired bundles carry both need-side and want-side polarities on different components are handled only conservatively under the current architecture: the full $\Delta \text{vol}_P(X)$ enters $\text{vol}_R^{\text{lower}}$ as want-side preference revelation, and the companion paper’s need-access cost diagnostic [Lasser, 2026d, Need Sufficiency] counts the entire mixed-event magnitude as need-access cost. A canonical instance is residential relocation, where the agent surrenders a bundle of capabilities (location, neighborhood, school district, commute) and acquires a different bundle of the same dimensions, with shelter as the need-side baseline and the remaining dimensions ranging from side-move to upgrade. From the trade ledger alone, a side-move-with-one-upgrade is currently indistinguishable from an upgrade-with-one-side-move; both register as the trade’s full $\Delta \text{vol}_P(X)$. An event-local bundle decomposition with polarity tags ([Lasser, 2026d, Need Sufficiency] flags this as the NAC-tightening direction) is the natural follow-up: $\Delta \text{vol}_P(X) = \sum_c \alpha_{n,c}^X \cdot \Delta \text{vol}_{P_c}(X)$ on the surrendered side with each $\alpha_{n,c}^X$ carrying a need-side / want-side polarity tag, yielding a need-share and a want-share separately.
3. **Non-stationarity of revealed preferences and trade-exercise temporal mismatch.** Preferences shift over time. The bound $\text{vol}_R^{\text{lower}}$ reported by the framework is the *window-active historical* aggregate: it summarises trade evidence over the trade window W . “Current” exercise at time t may differ from the historical aggregate when preferences shift mid-window or after W ends. The window-active framing preserves the formal interpretation (“ $\text{vol}_R^{[W]}$ ” is the latent quantity the bound targets, which captures any exercise during W ,

not exercise at a specific instant), but a downstream user who reads $\text{vol}_R^{\text{lower}}$ as a current diagnostic should account for the window’s age and any preference-shift signal. An EWMA decay paralleling the risk-trust dynamics of [Lasser, 2026g, Definition 4] seems natural but requires empirical calibration.

A related coverage limitation is the *trade-exercise temporal mismatch*: a sacrifice event X_n at time $t_n \in W$ may be made on the basis of anticipated exercise that does not materialize within W (e.g., a long-horizon acquisition whose exercise begins only after W ends). S0 in window-active form bounds $U_i(Y_n)$ by $\text{vol}_R^{[W]}$, so such trades technically violate S0 (agent valuation reflects future-window exercise that is invisible to the current window) and are filtered if S0-attestation is enforced. The aggregate is therefore biased toward agents whose exercise lands within W relative to those whose exercise lands beyond W : trade-early/ exercise-late patterns are systematically underrepresented. This is conservative (it shrinks $\text{vol}_R^{\text{lower}}$, never inflates it), but it should be acknowledged when interpreting low β^{lower} in populations with long-horizon trades. A multi-window or rolling-window extension that aggregates across overlapping W_k to capture deferred exercise is a natural follow-up; the formal extension is straightforward but the empirical calibration of how far to extend the rolling window is a deployment-dependent governance question.

4. **Coerced sacrifice.** Assumption S1 (free choice) is load-bearing. Under duress, the agent’s trade does not reveal preferences. Filtering coerced sacrifice events may require commitment-based attestation of choice sets [Lasser, 2026a, Exogenous Verification].
5. **Market-failure cases.** In contexts without functioning markets (pre-capitalist societies, closed economies, intra-family sharing), the money-sacrifice channel is quiet even though vol_R activity is real. Does the time-sacrifice channel alone suffice, or does the framework need a non-market sacrifice variant?
6. **Commitment infrastructure at scale.** A fully committed trade ledger at the scale of a modern economy is an infrastructure problem. The privacy guarantee (Section 6) is conditional on the commitment infrastructure existing, which is the remit of [Lasser, 2026a, Exogenous Verification].
7. **Population-level calibration (S0).** Assumption S0 requires that individual agents do not overvalue unbenchmarked bundles. A robust version of Theorem 2 would replace S0 with a weaker population-level condition: $E[U_i(Y)] \leq \text{vol}_R(Y) + \epsilon$ for some bounded bias ϵ . Specific behavioural mechanisms that stress individual-agent S0 and motivate a population-level weakening include: (i) *ritualized payment*, where a transaction’s nominal price is symbolic rather than exchange-valued (a token fee to “soften” receiving something for free); (ii) *gift aversion*, where agents refuse free offers because they interpret absence of price as signalling hidden defect, inflating observed sacrifices on nominally free goods; (iii) *time-for-money inefficiency*, where a given hour’s wage rate r_i does not reflect the agent’s marginal valuation (labour-market frictions, status wages, family-care default allocations). Each of these produces systematic biases on observed sacrifice magnitudes that the per-agent S0 cannot absorb; a population-level reformulation that lets such biases cancel (or bounds the residual) is the natural follow-up.

Further open questions in the companion paper. Open questions on threshold governance, need identification, capability-stuffing as structural avoidance, and the full architecture of gap-decomposition governance are collected in the companion paper [Lasser, 2026d, Need Sufficiency].

Author Contributions

Teague Lasser identified the B-to-C gap as a structural open problem in the GFM sequence and directed the reframing from direct vol_R measurement to revealed-sacrifice observation. Directed the formal development of the two-channel (money and time) sacrifice model, the aggregate lower bound theorem, and the commitment-layer composition. Identified the duality between controlled relaxation and revealed sacrifice (risk dimension versus value dimension) and the shared privacy discipline (resolution through frequency, not depth). Made final editorial decisions and is responsible for the paper’s intellectual direction and all claims made.

Claude Opus 4.7 (Anthropic) drafted the formal exposition: the revealed-sacrifice observation model with its three formal privacy properties and two institutional preconditions (Section 2), the aggregate B-to-C lower bound theorem with its two-level structure (Section 3), the two-channel sacrifice analysis (Section 4), the axiom-inheritance analysis and non-self-balancing corollary (Section 5), and the commitment-layer composition including the ZK-aggregation proposition (Section 6).

GPT 5.5 (OpenAI) served as cold technical reviewer across thirty review cycles, catching formal gaps across all sections—including the requirement for pairwise poset-independence (not merely set-disjointness) in the theorem’s aggregation step, the distinction between vol_R being non-invariant under capability removal and a measure-axiom failure in the non-self-balancing finding, the distinct-label scope of the commitment rollup, the full poset-boundary closure (cooperative + subsumption) required by the combined diagnostic, and the scope-mismatch between Part A and Part B monotonicity claims.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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A The Exercise Indicator (Supplementary)

A.1 Scope limitation

The exercise indicator is retained as a *supplementary* measurement tool, applicable only where interior access is legitimate—specifically, within the collective’s own operational perimeter where no privacy issue arises (e.g., the framework measuring the exercise status of its own internal components, or a cooperative measurement within a consenting sub-collective).

Definition 11 (Exercise Indicator, Internal Perimeter). *Let G be a capability collective operating within an agreed-upon observation perimeter $O \subseteq P$. We require O to be boundary-closed in the poset-independence sense of Definition 10(i): neither cooperative-composition nor subsumption links cross the $O/P \setminus O$ partition. Concretely, two conditions:*

(BC1) Cooperative closure. *For every cooperative capability $z \in P$, either all participants of z lie in O or none do (so cooperatives are not split across the boundary).*

(BC2) Subsumption closure. *No subsumption relation crosses the boundary: if $c \succeq c'$ in P (i.e., c subsumes c'), then either both $c, c' \in O$ or both $c, c' \in P \setminus O$.*

Together (BC1) and (BC2) make O and $P \setminus O$ poset-independent in the sense of Definition 10(i), which is what licenses additive composition across the boundary below. (BC1) alone—the weaker cooperative-only condition—is insufficient: subsumption edges crossing the boundary would break M4 on the $O/P \setminus O$ partition.

For each capability $d \in O$, define:

$$e_t(d) = \begin{cases} 1 & \text{if there exists a realized cooperative output in } [t - \Delta, t] \cap W \\ & \text{such that removing } d \text{ from } O \text{ makes the output unrealizable,} \\ 0 & \text{otherwise.} \end{cases}$$

The lookback is clipped to the window W : for t near the left endpoint t_0 of W , the lookback interval is $[\max(t_0, t - \Delta), t]$, so realized outputs strictly before W do not contribute to $\text{vol}_R^{[W]}$. The indicator is scoped to the observation perimeter O . The framework may use $e_t(d)$ for internal diagnostics within O , but may not apply it to capabilities outside O (where privacy constraints apply).

The exercise-based realized volume within O at time t is the vol_P -weighted sum of currently-exercised capabilities in the perimeter:

$$\text{vol}_R^{\text{exact}}(O) := \text{vol}_P(\{d \in O : e_t(d) = 1\}).$$

For the window-active form used by Theorem 2, we set

$$\text{vol}_R^{\text{exact}[W]}(O) := \text{vol}_P(O \cap P_W^{\text{ex}})$$

where $P_W^{\text{ex}} = \bigcup_{t \in W} \{d \in O : e_t(d) = 1\}$. The instantaneous form $\text{vol}_R^{\text{exact}}(O)$ is the t -indexed specialization (right-endpoint t of the lookback); the window-active form $\text{vol}_R^{\text{exact}[W]}(O)$ is the union over $t \in W$. Bare $\text{vol}_R^{\text{exact}}(O)$ in body prose refers to the window-active form unless an instant t is explicit in the surrounding context, matching the same convention as Definition 3.

Remark 22 (The two channels are complementary). *Within the perimeter O , the exercise indicator provides exact exercise status ($e_t(d) \in \{0, 1\}$). Outside O , the sacrifice channel provides a lower bound. Under the boundary-closed hypothesis of Definition 11 and under Theorem 2 Part B hypotheses (so that the subset-level quantity $\text{vol}_R^{\text{lower}}(P \setminus O)$ is well-defined per Definition 5), the aggregate vol_R diagnostic is a valid lower bound obtained by additive composition:*

$$\text{vol}_R^{\text{combined}} := \text{vol}_R^{\text{exact}}(O) + \text{vol}_R^{\text{lower}}(P \setminus O) \leq \text{vol}_R(O) + \text{vol}_R(P \setminus O) = \text{vol}_R(P),$$

where the final equality uses axiom M4 applied to the poset-independent pair $O, P \setminus O$ (Section 5). $\text{vol}_R^{\text{exact}}(O)$ uses the exercise indicator and $\text{vol}_R^{\text{lower}}(P \setminus O)$ uses the sacrifice channel. Under Part A alone, $\text{vol}_R^{\text{lower}}$ is defined only at the per-bundle level; in that regime $\text{vol}_R^{\text{combined}}$ can be reported as the pair $(\text{vol}_R^{\text{exact}}(O), \text{vol}_R^{\text{lower}}_A)$ where $\text{vol}_R^{\text{lower}}_A$ is the Part A bundle-level lower bound restricted to events whose bundles lie in $P \setminus O$, without the subset-aggregation interpretation.

Remark 23 (When boundary-closure fails). *If O fails either condition—a cooperative z has participants split across O and $P \setminus O$ (BC1 fails), or a subsumption edge $c \succeq c'$ crosses the boundary (BC2 fails)—then O and $P \setminus O$ are not poset-independent, and M4 additivity across the boundary does not apply. In that regime, the compositional lower-bound interpretation of $\text{vol}_R^{\text{combined}}$ is not established: the paper proves $\text{vol}_R^{\text{combined}} \leq \text{vol}_R(P)$ only under BC1 + BC2. Outside that regime, the identity above becomes an equality $\text{vol}_R(O) + \text{vol}_R(P \setminus O)$ that no longer equals $\text{vol}_R(P)$, and no general inequality between $\text{vol}_R^{\text{combined}}$ and $\text{vol}_R(P)$ is available in this paper without an additional lemma on subsumption-crossing or cooperative-crossing decompositions of vol_P . $\text{vol}_R^{\text{combined}}$ can still be computed and reported as a heuristic composite of the two sub-diagnostics, but it should not be interpreted as a certified lower bound on $\text{vol}_R(P)$ in that case. Under BC1 failure specifically, the qualitative intuition is that z contributes to $\text{vol}_R(P)$ but to neither $\text{vol}_R^{\text{exact}}(O)$ nor $\text{vol}_R^{\text{lower}}(P \setminus O)$ individually—so $\text{vol}_R^{\text{combined}}$ would undercount the boundary cooperative’s vol_R mass if a lower-bound interpretation were otherwise available; under BC2 failure, the subsumption coupling makes the sign of the gap between $\text{vol}_R^{\text{combined}}$ and $\text{vol}_R(P)$ undetermined by this paper’s machinery.*

Practical perimeters (the collective’s own operational scope, consenting sub-collectives) typically satisfy both boundary-closure conditions because cooperative capabilities and subsumption hierarchies are either wholly internal to the perimeter or wholly external. When designing the perimeter, choose O to respect both the cooperative and subsumption structure of P ; the combined diagnostic’s compositional guarantee requires both.

B Proof Sketches and Technical Notes

B.1 Privacy-minimality (Section 2)

Theorem 1 + Proposition 1 strategy. The three formal properties (P1, P3, P4) are proved independently from the channel’s construction; the two institutional preconditions (P2, P5) are documented separately in Proposition 1. Proof strategy: P1 (no interior access): by construction of the observation tuple. P3 (discretization): by bounding channel bandwidth with trade frequency. P4 (commitment composability): by composition with [Lasser, 2026a, Definition 6]. P2 and P5 are not theorem content: P2’s zero-contribution property is formal (a consequence of the aggregate’s summation structure), but the consent interpretation depends on institutional opt-in conventions; P5 depends on public-ledger infrastructure external to the paper’s formal scope, and is additionally split between money-channel events (observable from public ledger) and time-channel events (require an additional public wage schedule or imputation rule).

B.2 Aggregate lower bound (Section 3)

Theorem 2 strategy. *Part A (bundle-level, S0–S2 and S4(a), poset-independent bundles, genuine-exchange conditions):*

1. Revealed-preference observation: $U_{i_n}((H_n \setminus X_n) \cup Y_n) \geq U_{i_n}(H_n)$.
2. Cancellation to bundle comparison (S4(a) + Definition 2 conditions (i)–(iv), via Lemma 1): $U_{i_n}(Y_n) \geq U_{i_n}(X_n)$.
3. Calibration chain (S0 + S2), S0 stated directly in window-active form: $\text{vol}_R^{[W]}(Y_n) \geq U_{i_n}(Y_n) \geq U_{i_n}(X_n) \geq \Delta \text{vol}_P(X_n)$. S0’s consequent is against $\text{vol}_R^{[W]}$ rather than against the instantaneous $\text{vol}_{R_{t_n}}$: a sacrifice event records acquisition, not an instantaneous exercise witness, so the bound is about cumulative exercise of Y_n anywhere in W (Definition 3, Section 2.1).
4. Poset-independence of bundles (Definition 10) gives M4 additivity (inheritance proved in Theorem 3 via sub-poset restriction): $\text{vol}_R^{[W]}(\bigcup Y_n) = \sum_n \text{vol}_R^{[W]}(Y_n) \geq \sum_n \Delta \text{vol}_P(X_n)$.

Part B (per-capability refinement, additionally S3–S5 and within-bundle poset-independence; conditional on S5, see Remark 1):

4. Part B is stated for events with $\Delta \text{vol}_P(X_n) > 0$ and bundles Y_n satisfying within-bundle poset-independence (Definition 4, Remark 9); bundles with internal cooperative activation fall back to Part A only. Positivity $\text{vol}_R(Y_n) > 0$ then follows from Part A.
5. Within-bundle poset-independence plus M4 gives $\text{vol}_R(Y_n) = \sum_{c \in Y_n} \text{vol}_R(c)$, making the partition-of-unity constraint $\sum_{c \in Y_n} \alpha_{n,c} = 1$ compatible with S5’s share inequality $\alpha_{n,c} \leq \text{vol}_R(c)/\text{vol}_R(Y_n)$ (equality achievable). Combined: $\text{vol}_R(c) \geq \alpha_{n,c} \cdot \Delta \text{vol}_P(X_n)$. S3 (bundle coherence) ensures the hedonic decomposition is non-degenerate so $\alpha_{n,c}$ is well-specified.
6. Per-capability max-attribution across overlapping events (Proposition 2).
7. Summation over capabilities in U (pairwise poset-independent) via a second M4 application.

Key technical challenges:

- *Poset-independence (Part A):* the category partition $\mathcal{C}$ (Definition 10) must be a genuine poset partition. Set-disjointness is necessary but not sufficient (Remark 11).
- *Decomposition validity (S5, Part B):* hedonic regression [Rosen, 1974] produces attribution weights, not vol_R share lower bounds. S5 is the explicit bridge.
- *Calibration (S0):* the substantive bridge of the chain (Remark 2)—a one-sided conservatism claim whose empirical verification is the hardest among the assumption stack (Open Question 7).

B.3 Axiom inheritance (Section 5)

Theorem 3 strategy. Axiom-by-axiom analysis: M1–M5 are direct from properties of summation/max over non-negative terms (detailed in the body; M4 is proved constructively via subposet restriction). M6 for vol_R as a measure on a fixed exercised sub-poset is inherited from vol_P by restriction—no axiom failure there. The paper’s non-self-balancing claim rests instead on *capability-removal non-invariance*: removing an unexercised capability leaves vol_R unchanged while vol_P drops by the removed weight (Example 3). This is a system-level property (of the mapping $\mathcal{P} \mapsto \text{vol}_R^{\mathcal{P}}$), not a measure-axiom failure. M6(b) for $\text{vol}_R^{\text{lower}}$ is witnessed separately by Example 4: max-attribution subadditivity. Corollary 2 follows from (a)’s non-invariance, not from (b)—a lower bound failing measure axioms is unremarkable; what breaks self-balancing transfer is that vol_R doesn’t penalize shedding of unused capabilities.

C Notation Summary

C.1 Symbols introduced in this paper

Symbol	Definition	Ref.
(i, X, Y, t)	Revealed-sacrifice event	Def. 2
$\Delta \text{vol}_P(X)$	vol_P contribution of surrendered capability	Def. 2
$\text{vol}_R^{\text{lower}}(Y)$	Lower bound on vol_R of acquired bundle	Def. 2
$[t_0, t_0 + T]$	Aggregate trade window	Def. 1
$P_t^{\text{ex}}, P_W^{\text{ex}}$	Exercised sub-poset (instantaneous / window-active)	Def. 3
$\text{vol}_{Rt}, \text{vol}_R^{[W]}$	Realized capability volume (instantaneous / window-active)	Def. 3
$\alpha_{n,c}$	Event-local decomposition coefficients	Def. 4
$\text{vol}_R^{\text{lower}^A}(E)$	Part A lower bound via sub-batch supremum	Prop. 3
$\beta^{\text{lower}}(G, T)$	B-to-C ratio under revealed sacrifice	Def. 6
p	Price (money-sacrifice signal)	Def. 7
r_i	Agent i 's market wage rate	Def. 8
(Γ, π, t)	Committed revealed-sacrifice event	Def. 9
$\mathcal{C}$	Public poset-independent partition of U	Def. 10
$\text{category}(Y)$	Bundle's partition label	Def. 10
$e_t(d)$	Exercise indicator (internal perimeter)	Def. 11
O	Observation perimeter	Def. 11
$\text{vol}_R^{\text{exact}}(O)$	Exercise-based realized volume within O	Def. 11
$\text{vol}_R^{\text{combined}}$	Combined diagnostic (exact + lower bound)	App. A

C.2 Assumptions

Assumption	Name	ZK-verifiable?
S0	Valuation bounded by exercise ($U_i(Y) \leq \text{vol}_R^{[W]}(Y)$)	No
S1	Free choice (non-degrading alternative)	No (default)
S2	Benchmark grounding ($U_i(X) \geq \Delta \text{vol}_P(X)$)	Objective part only
S3	Bundle coherence (non-degenerate decomposition)	No (default); Yes with richer commitment
S4(a)	Valuation additivity	No
S4(b)	vol_R additivity on poset-independent capabilities	Yes (via poset)
S5	Decomposition validity (α are share lower bounds)	No

C.3 Dependencies on prior papers

Paper	Result	Number	Used in
[Lasser, 2026b, Foundation]	Population empowerment (vol_P definition)	Def. 6	§1
[Lasser, 2026f, Poset Definition]	Benchmark	Def. 2	Def. 2
[Lasser, 2026f, Poset Definition]	Poset measure vol_P	Def. 7	Def. 2, Def. 6
[Lasser, 2026f, Poset Definition]	Axioms M1–M6 (vol_P axiomatization)	Prop. 1	§1, Thm. 3
[Lasser, 2026f, Poset Definition]	Self-balancing on posets	Thm. 1	§1, Cor. 2
[Lasser, 2026c, Horizon Aware]	Anti-monopolar property	Prop. 6	§1
[Lasser, 2026c, Horizon Aware]	Preemptive-restriction criterion	Prop. 1	§7
[Lasser, 2026g, SCM]	Observation-window framing	(general)	Def. 1
[Lasser, 2026a, Exogenous Verification]	Commitment Protocol	Def. 6	Def. 9
[Lasser, 2026a, Exogenous Verification]	ZK Capability Proof	Def. 7	Def. 9
[Lasser, 2026e, Phase Redundancy]	Phase boundary	Thm. 1	§7
[Lasser, 2026e, Phase Redundancy]	Channel redundancy	Thm. 2	§7
[Lasser, 2026h, Controlled Relaxation]	Controlled relaxation	Def. 3	§7
[Lasser, 2026h, Controlled Relaxation]	Damage bound	Thm. 1	§7
[Lasser, 2026d, Need Sufficiency]	Need-sufficiency architecture (S1 filter)	§2	Sec. 2 (as external precondition)
[Lasser, 2026d, Need Sufficiency]	Gap decomposition	§3	§7
[Lasser, 2026d, Need Sufficiency]	Wireheading-consistent signature	§4	§7
[Lasser, 2026d, Need Sufficiency]	Residual class characterization	§5	§7